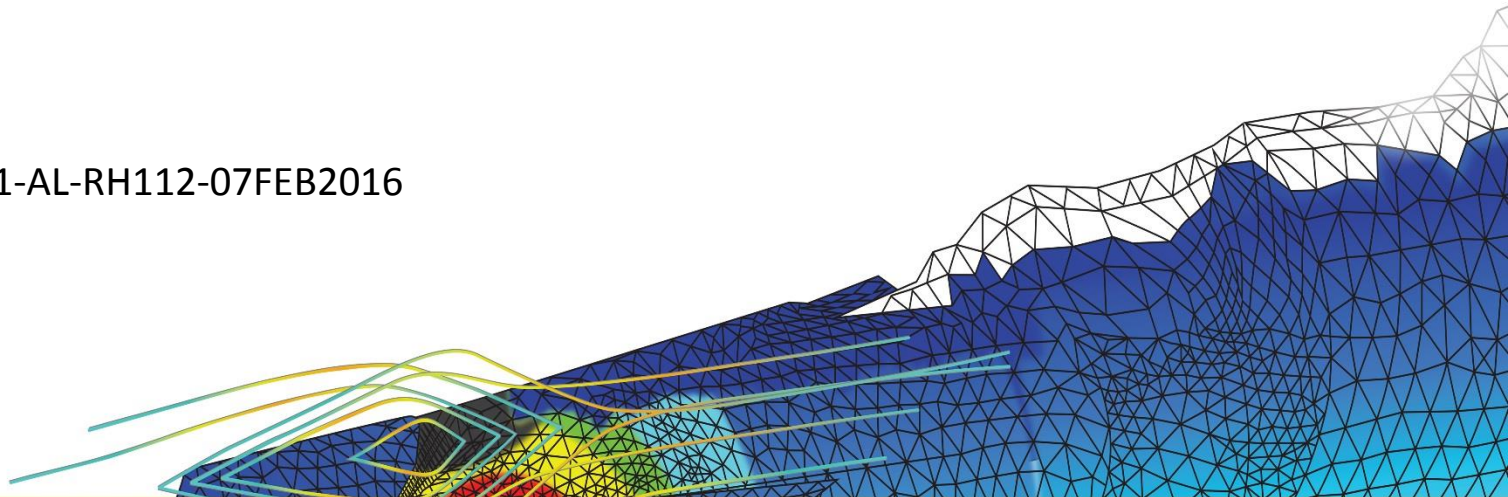




RedHawk™ Vectorless Dynamic Analysis

VERSION: V5.1-AL-RH112-07FEB2016

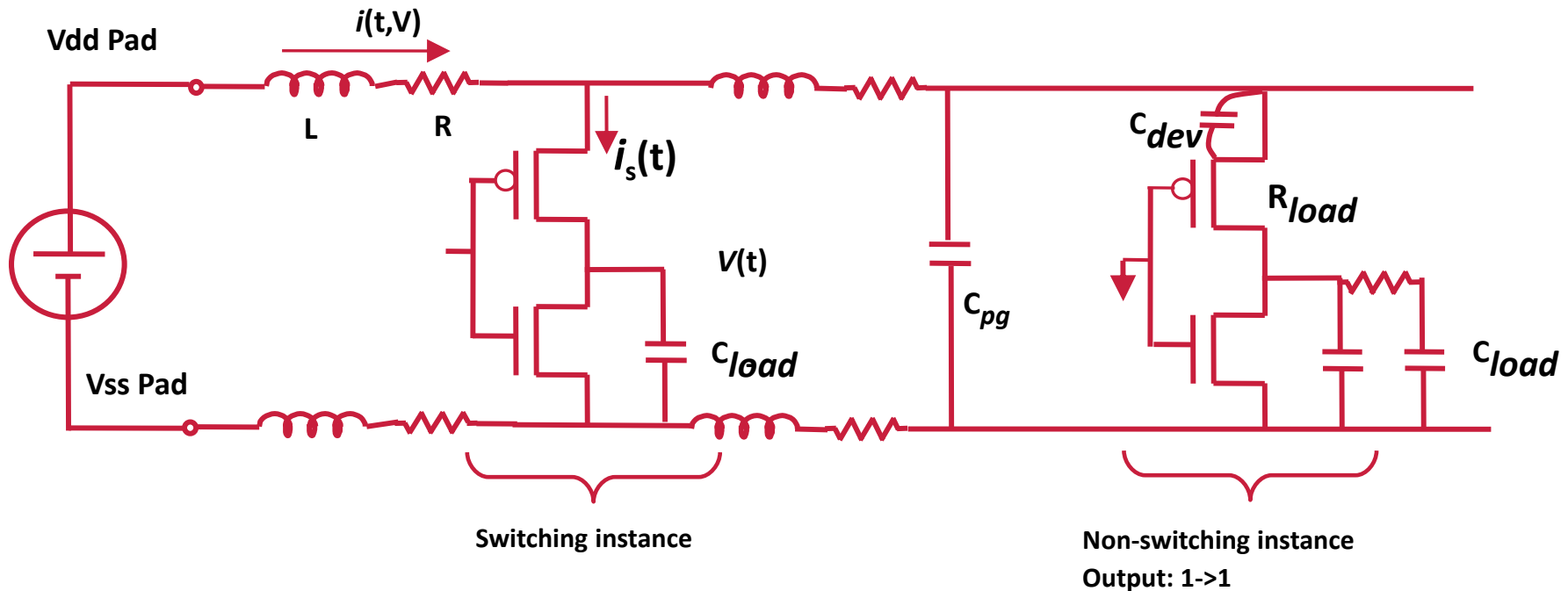


Agenda

- Theory
- RedHawk Dynamic flow
- Analysis of results
- Conclusions

Dynamic Voltage Drop Problem Definition

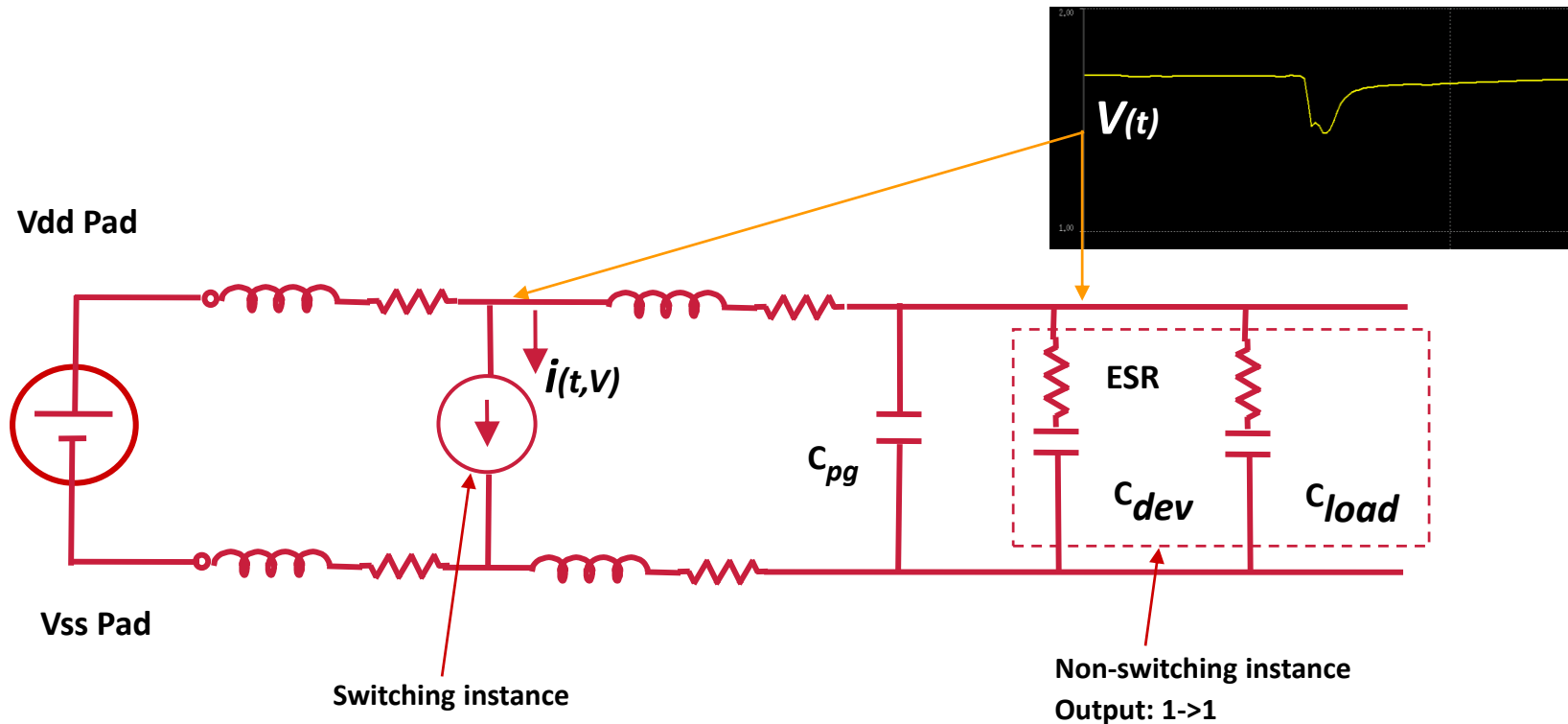
On-chip power/ground network



- On-chip power/ground network → R,L,C mesh
- Switching instances → $i(t, V)$ sources
- Non-switching instances → Effective decaps, ESR and leakage

Dynamic Voltage Drop on P/G Network

- PWL current for each instance
- V_{dynamic} waveform is computed at every node by transient simulation



Dynamic Analysis Which, When, and What

- **Logic: will an instance switch?**
 - VCD mode: switching pattern is provided by user
 - Vectorless mode: Constraint-based statistical switching
- **Timing: if it switches, when will it switch?**
 - STA-based timing window, slew
- **Power: if it switches, what is its current profile?**
 - Spice-based library characterization (APL)

Vectorless Dynamic Detail

Logic Aware

- Consideration of cell functionality
- Consideration of instance specific toggle
- Propagation of state/probability

Power Aware

- Higher switching probability for instances with more energy
- Statistical determination to limit artificial hot-spots

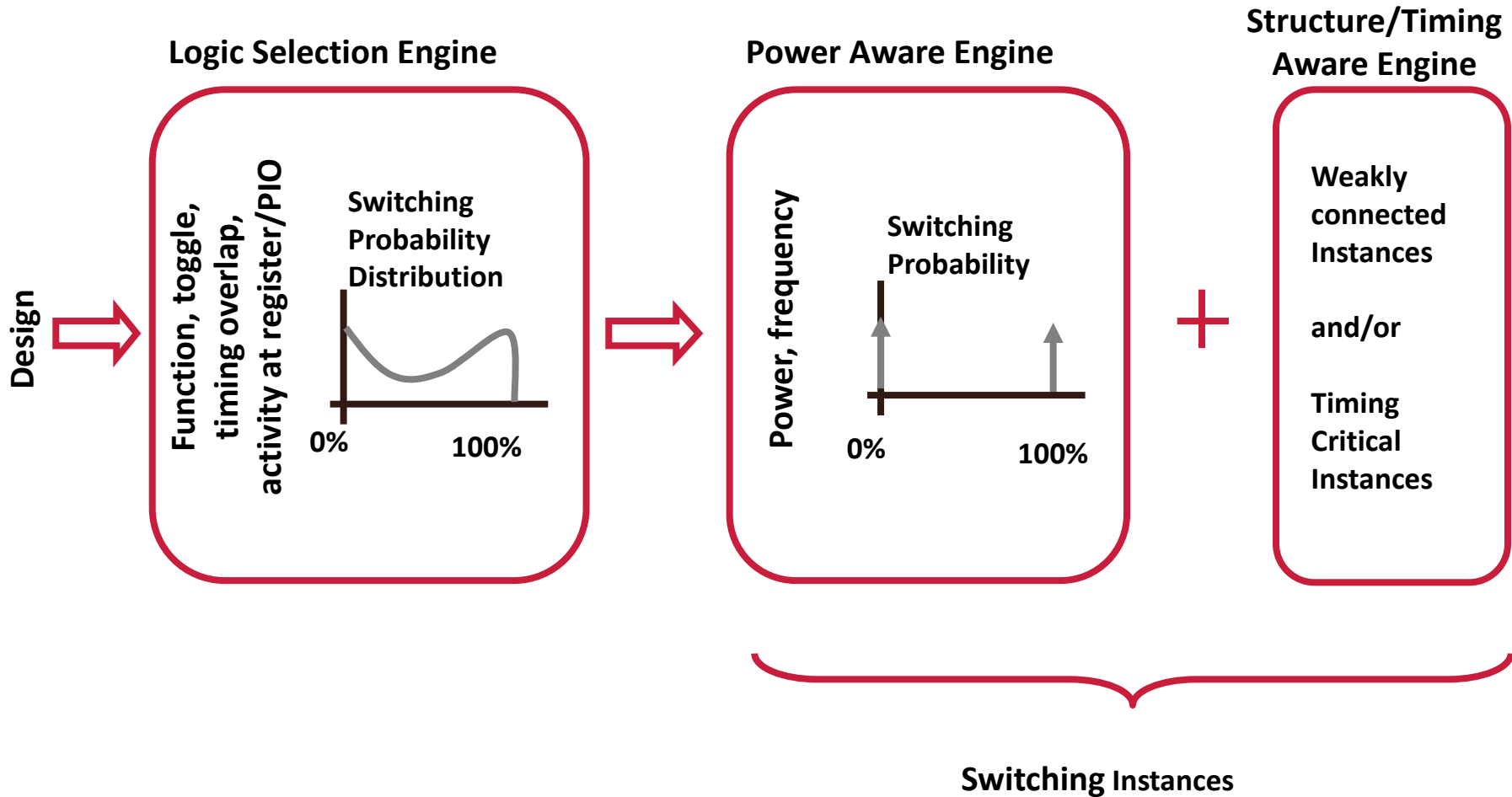
PG Aware

- Understanding of P/G network structural weaknesses
- Switching biased towards highlighting areas of poor connections

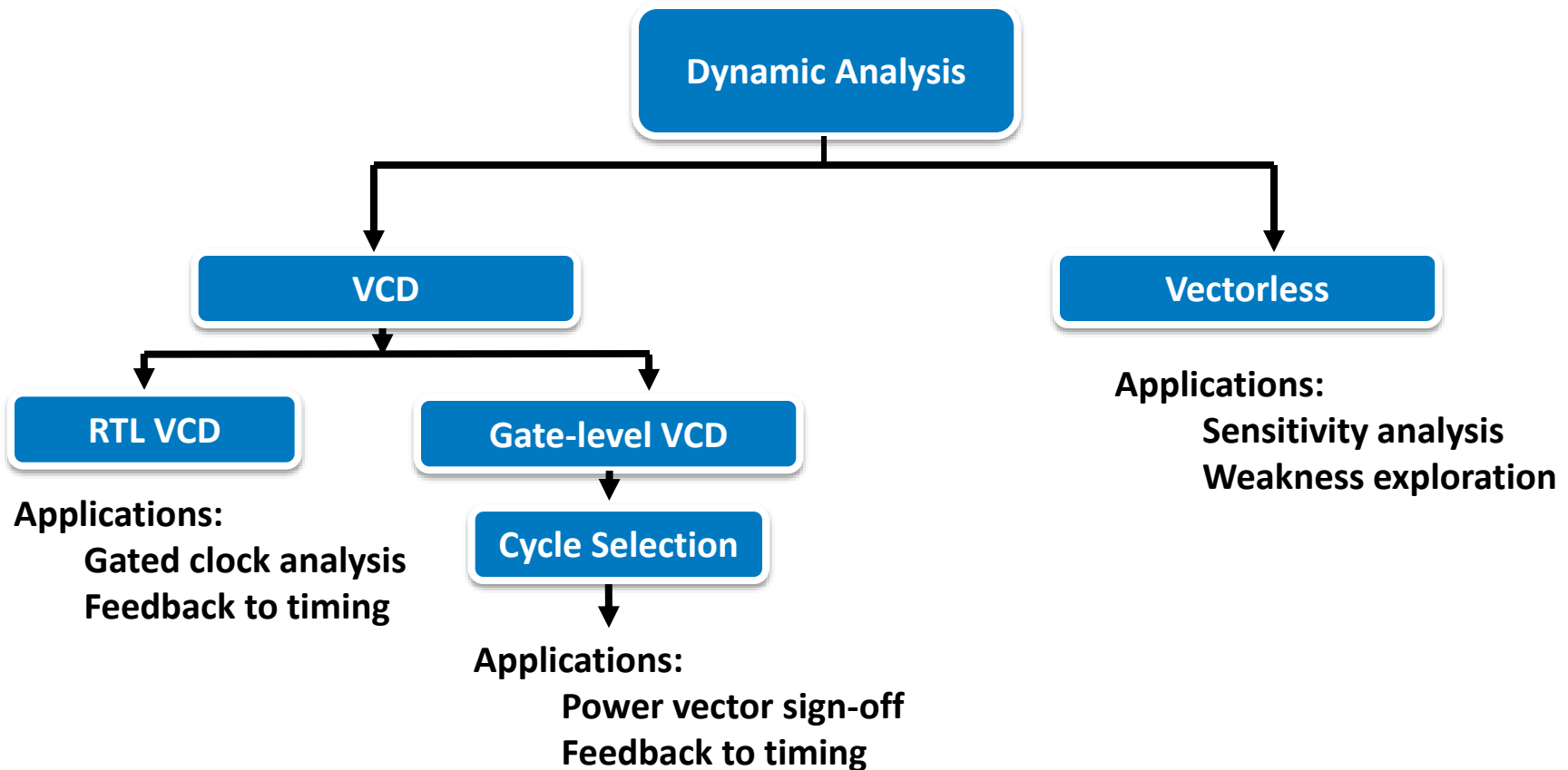
Timing Aware

- Consideration of timing window overlap
- Switching biased towards user specified critical paths
- Instances in the paths, instances that affect the critical paths

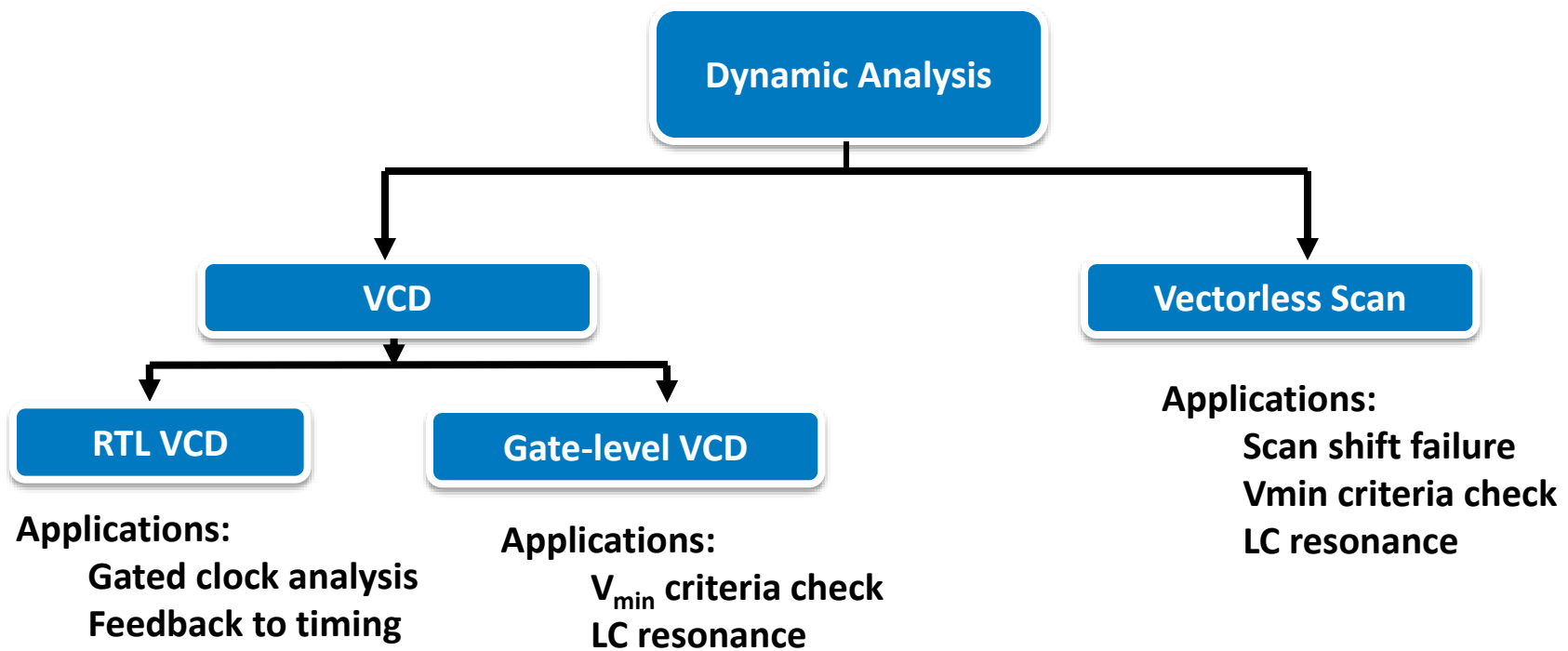
RedHawk VectorLess Analysis



Dynamic Power Integrity Flow Methodology (Functional Mode)



Dynamic Power Integrity Flow Methodology (Scan Mode)

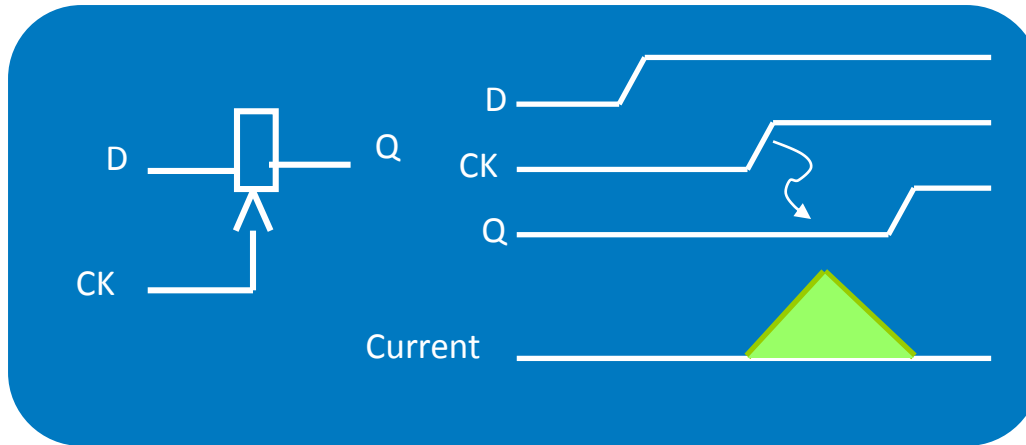


VCD-based Analysis

- **VCD**
 - User specified switching interval (frame) simulated
 - RH vcdscan utility analyzes VCD and reports intervals with highest power
- **VCD-guided vectorless**
 - Uses toggle activity from VCD (entire or frame specific) to guide switching scenario creation
- **Supports RTL VCD and state propagation**

Enabling Technologies Apache Power Library

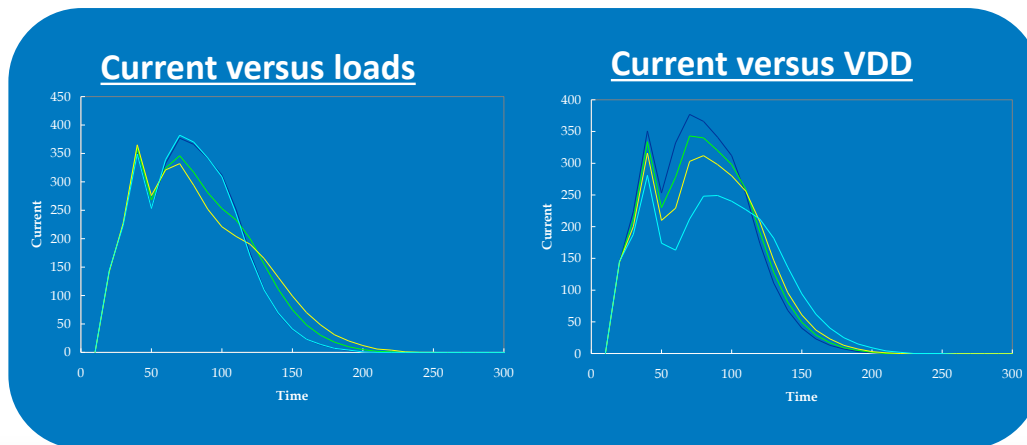
.Lib: Triangular energy waveforms



APL model for a cell

- Cell level abstraction with transistor level characterization
- For standard cells, memories and macros

APL: Multi-dimensional current profiles



Contains:

- Switching current
- Leakage $\sim f(\text{state, temperature})$
- Intrinsic capacitance
- Effective series resistance
- Cell delays and slews

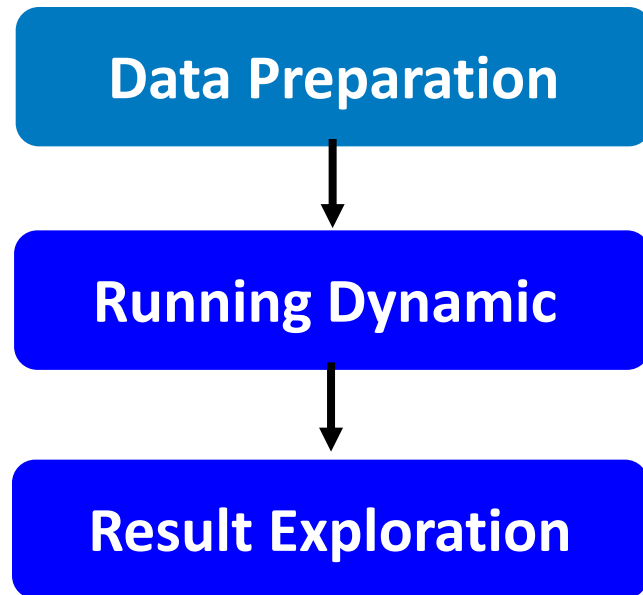
...

Dynamic Analysis Using .LIB vs APL

- .lib profiles have a single peak only and cannot correctly capture multi-stage cell switching currents
- .lib profiles cannot capture the change of operating voltage for cells
- RedHawk using APL can adjust the switching current of cell based on its VDD and VSS voltage values

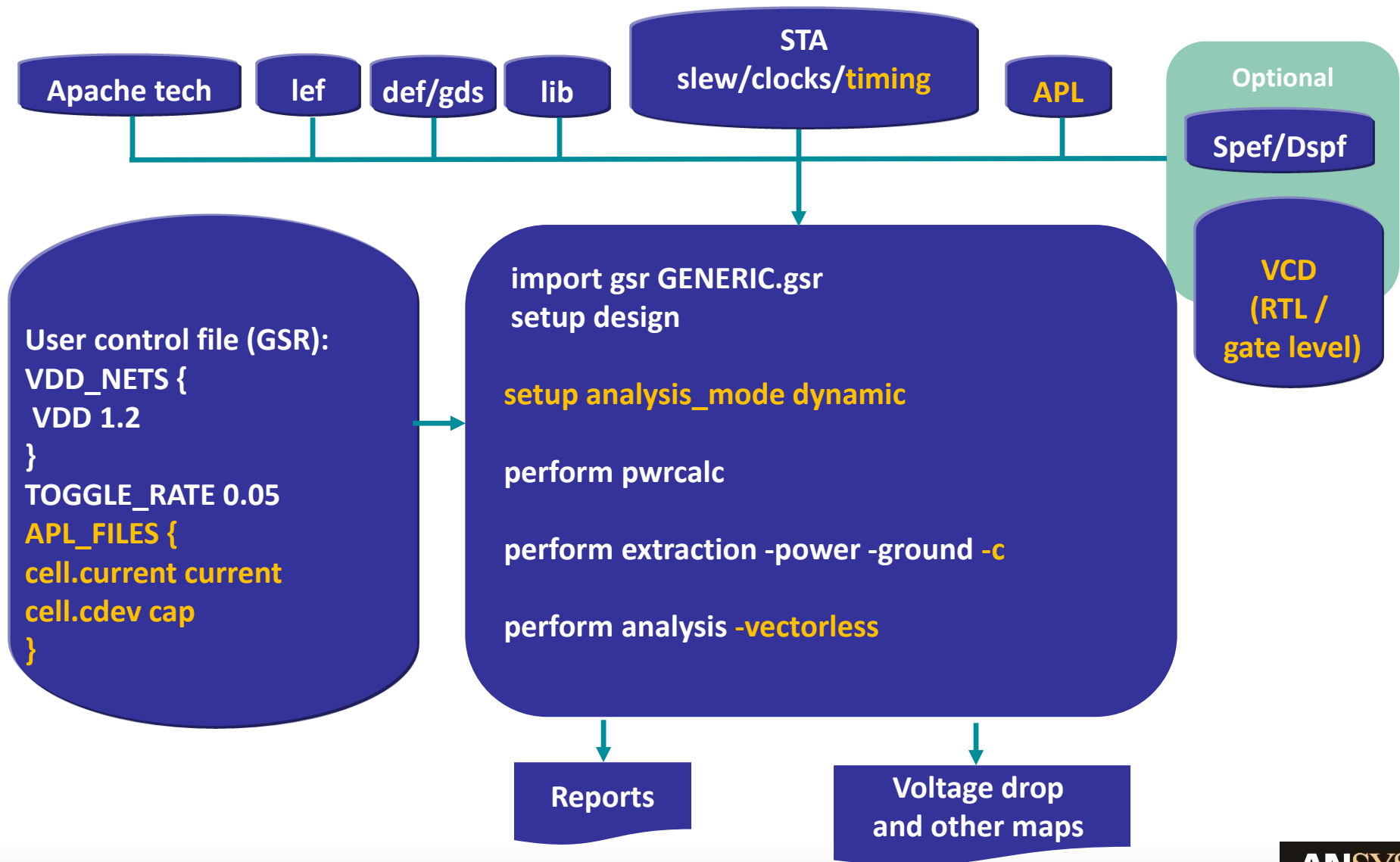
	APL based	.LIB based
Switching Current Profile	True spice current waveform	Triangular profile using .lib charge and delay
Intrinsic Decap and effective series resistance	True spice characterized value	None used
Leakage Current		.lib based value

Data Preparation for Dynamic



- Setup dynamic run
- Reviewing the additional GSR keywords and command file changes

Dynamic Analysis Flow



Setting Up The Dynamic Run

- **cd GENERIC_tutorial/dynamic_run**
- **Review the GSR and Command file**
- **Kickoff the dynamic run**
 - `redhawk -f run_dynamic.tcl`

GSR Options for Dynamic Analysis

Dynamic simulation time determines length of transient simulation
#(default: 1/Freq)

DYNAMIC_SIMULATION_TIME 2.56e-9

Turn on dynamic pre-simulation

specify time or -1 for automatic setting

The second entry is the time-step speed-up during

pre-simulation (default = 1, same as the simulation time-step)

DYNAMIC_PRESIM_TIME -1 2

Transient simulation time step (default: 10ps)

DYNAMIC_TIME_STEP 25e -12

adsRpt/power_summary.rpt

Recommended dynamic simulation time, 2560psec ,to include 95.1044%
of total power for DYNAMIC_SIMULATION_TIME in GSR

Dynamic Run Command File

Data Preparation



Running Dynamic



Result Exploration

```
# Import data
import gsr GENERIC.gsr
setup design

setup analysis_mode dynamic

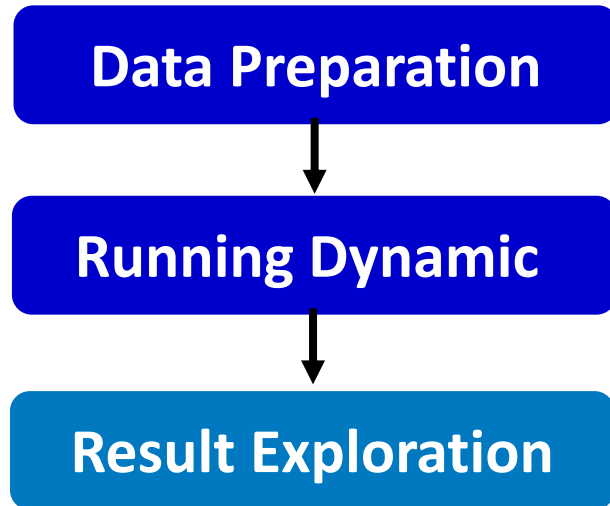
# Calculate power
perform pwrcalc

# Power/Ground grid extraction
# (extract capacitance too!)
perform extraction -power -ground -c

# Lumped resistance (in Ohms)
# for package, wirebond and pads
setup package -power -r 0.005 -l 2.5 -c 5
setup package -ground -r 0.005 -l 2.5 -c 5
setup wirebond -power -r 0.01 -l 2.2 -c 1.42
setup wirebond -ground -r 0.05 -l 1.7 -c 0.2
setup pad -power -r 0.001
setup pad -ground -r 0.001

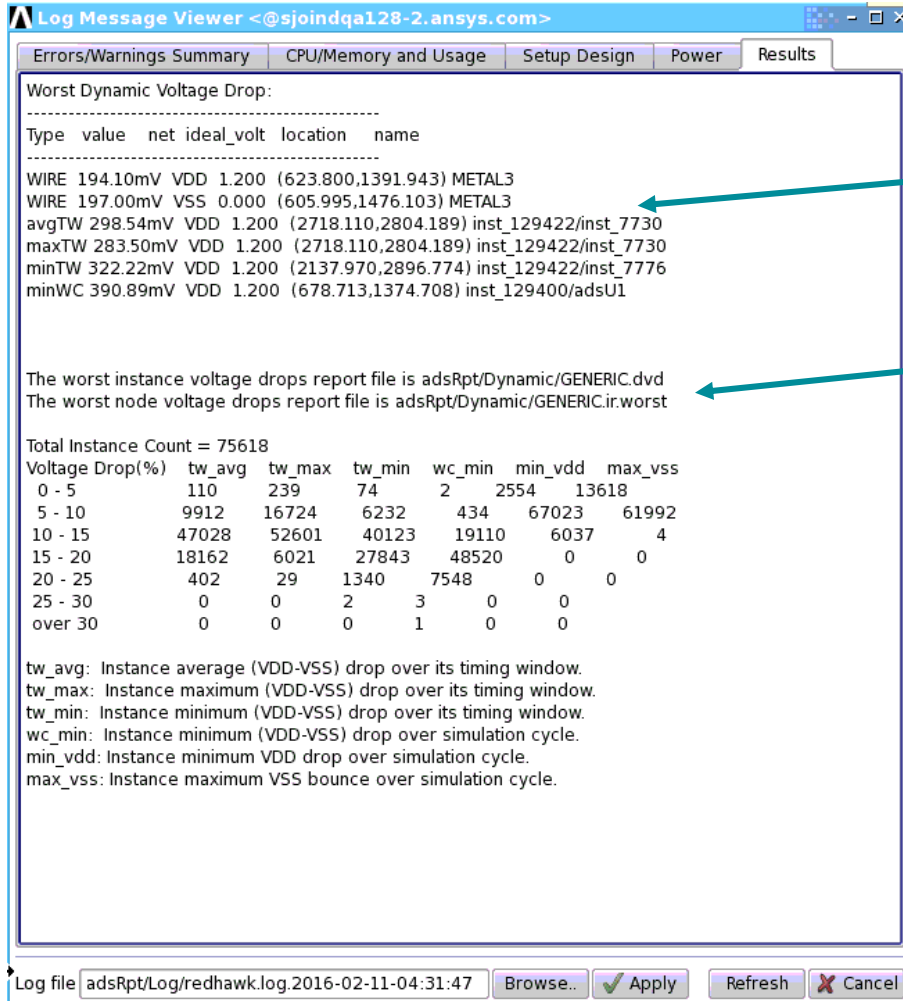
# Dynamic IR analysis
perform analysis -vectorless
explore design
```

Result Exploration and Debugging



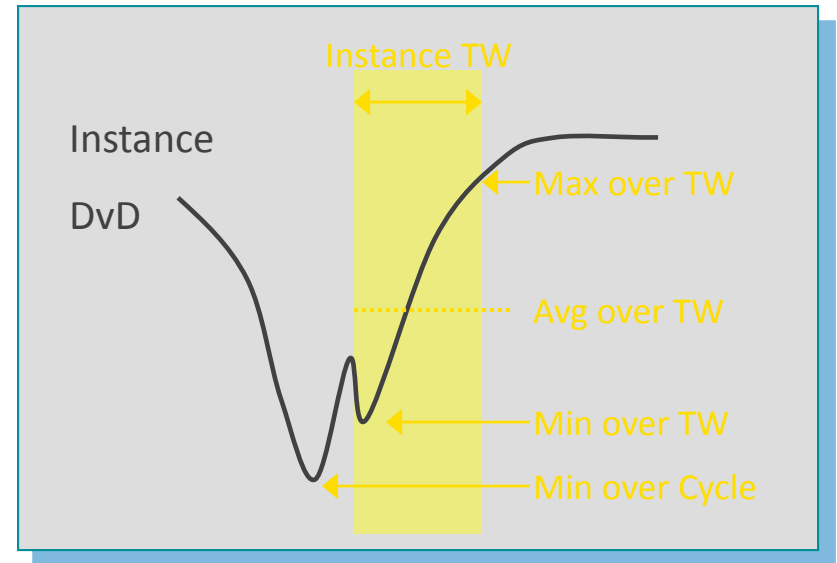
- Log message viewer
- Voltage Drop Maps
- Voltage and current w/f
- Dynamic Movie
- Design Report
- RedHawk Explorer

Examine Summary of Results



DVD summary

Text report pointers



* Log/error/warnings/command history stored under adsRpt/<>.date

Examine Summary of Results (Cont'd)

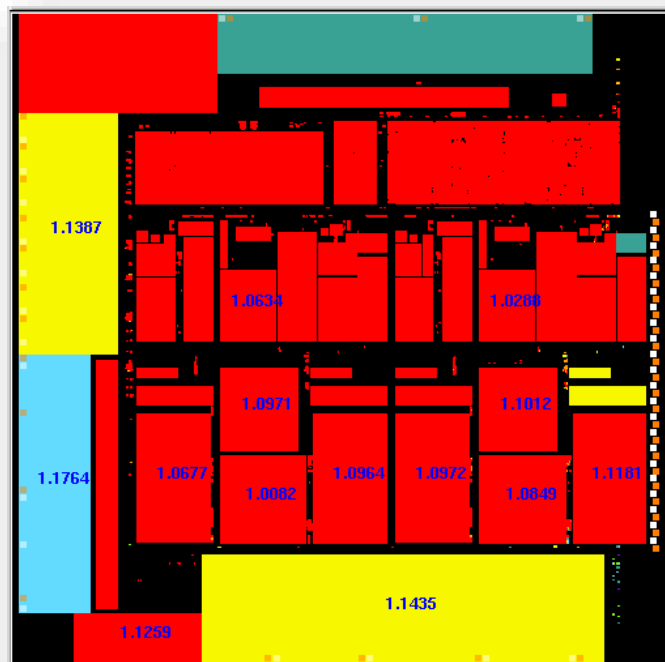
- Summarizes variation in current load for all switches
- Can identify the in-effective switches from this report
- Switches supplying high current also cause high voltage drop

```
#Report dynamic results of switch voltage and current
#instance_name type worst_int_Vdd/Vss(Volt) maximal_Vsw(Volt) maximal_Isw(Amp)
(SAT)
#SAT: maximal_Isw > saturation_current as specified in switch model file
inst_129973/switch_inst_R17_C52 header 1.119756e+00 1.922202e-02 9.611333e-04
inst_129973/switch_inst_R16_C52 header 1.119146e+00 1.525557e-02 7.628108e-04
inst_129973/switch_inst_R15_C52 header 1.118383e+00 1.235688e-02 6.178761e-04
inst_129973/switch_inst_R14_C52 header 1.117773e+00 1.022124e-02 5.110944e-04
inst_129973/switch_inst_R13_C52 header 1.117010e+00 8.848190e-03 4.424418e-04
inst_129973/switch_inst_R12_C52 header 1.116247e+00 7.932901e-03 3.966773e-04
inst_129973/switch_inst_R11_C52 header 1.115789e+00 7.475138e-03 3.737891e-04
inst_129973/switch_inst_R10_C52 header 1.115332e+00 7.475257e-03 3.737951e-04
inst_129973/switch_inst_R9_C52 header 1.115332e+00 7.475257e-03 3.737951e-04
inst_129973/switch_inst_R8_C52 header 1.115484e+00 7.932901e-03 3.966773e-04
inst_129973/switch_inst_R7_C52 header 1.116095e+00 8.695722e-03 4.348183e-04
inst_129973/switch_inst_R6_C52 header 1.116857e+00 9.458423e-03 4.729534e-04
inst_129973/switch_inst_R5_C52 header 1.117925e+00 1.067889e-02 5.339766e-04
inst_129973/switch_inst_R4_C52 header 1.119603e+00 1.220441e-02 6.102527e-04
```

“switch_dynamic.rpt” stored under: adsRpt/Dynamic

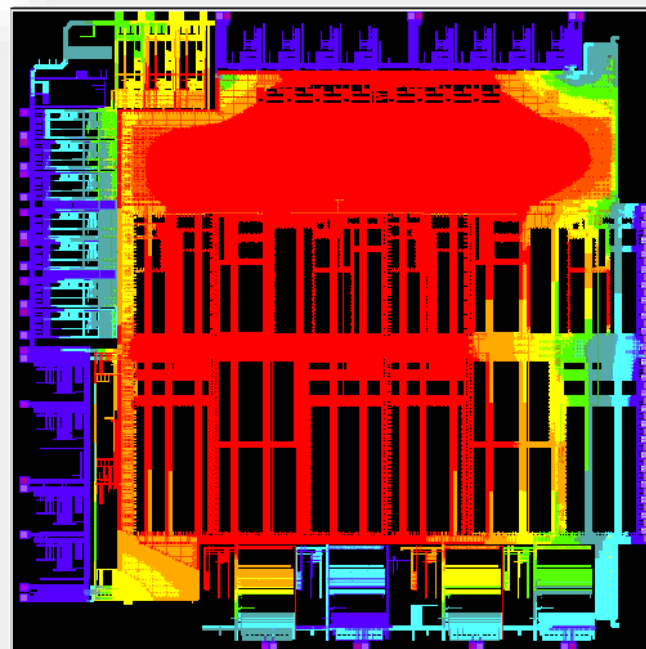
Explore Dynamic Results

- **View → Dynamic Instance DvD**
 - Max Vdd-Vss Over Time Window
 - Min Vdd-Vss Over Time Window
 - Avg Vdd-Vss Over Time Window
 - Min Vdd-Vss Over Whole Cycle



Avg Vdd-Vss Over Time Window

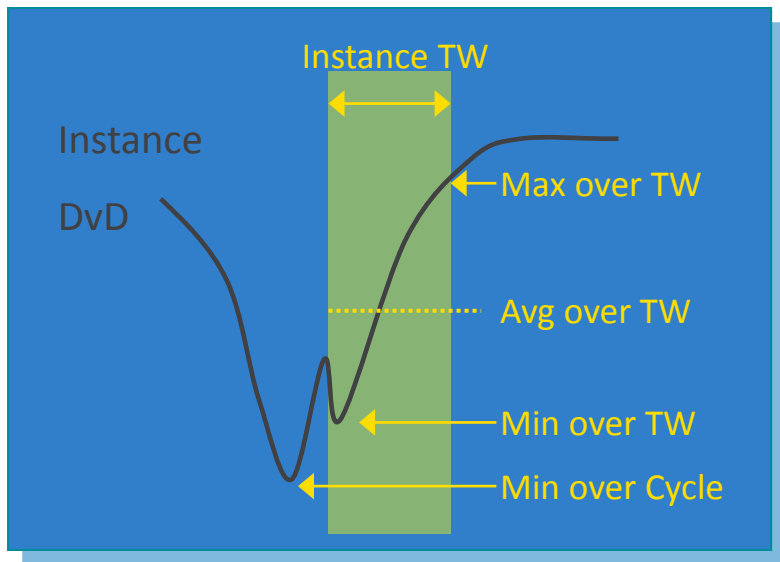
- **View → Voltage Drop Maps**
 - Wire & Via
 - Instance



Voltage Drop Map : Wire&Via

Explore Dynamic Results (Cont'd)

- Worst Instance DvD report
- Results -> List of Worst Instance DVD
- Avg/Max/Min over TW and Min Whole Cycle
- Need 'DYNAMIC_SAVE_WAVEFORM 1' to use 'DvD Plot' (disk space!)



List of Worst instance DVD <@sjoindqa128-2.ansys.com>

	Ideal Vdd	Avg DV	Max DV	Min DV	Min DV WC	Location(x y)	Name
1	1.2000	0.9015	0.9165	0.8931	0.8931	2718.11 2804...	inst_129422/inst_7730
2	1.2000	0.9421	1.0050	0.8778	0.8507	2137.97 289...	inst_129422/inst_7776
3	1.2000	0.9453	1.0002	0.9453	0.9453	2683.41 388...	inst_129974/inst_492109
4	1.2000	0.9454	0.9631	0.9440	0.9440	2617.63 3895...	inst_129974/inst_492042
5	1.2000	0.9456	1.0000	0.9442	0.9442	2633.73 389...	inst_129974/inst_492036
6	1.2000	0.9458	1.0230	0.9450	0.9450	2591.41 3880...	inst_129974/inst_491655
7	1.2000	0.9458	0.9961	0.9447	0.9447	2648.68 385...	inst_129974/inst_495145
8	1.2000	0.9458	0.9641	0.9448	0.9448	2629.36 386...	inst_129974/inst_491807
9	1.2000	0.9459	0.9538	0.9450	0.9450	2591.41 3884...	inst_129974/inst_492082
10	1.2000	0.9459	0.9572	0.9453	0.9453	2679.27 387...	inst_129974/inst_492104
11	1.2000	0.9460	1.0226	0.9453	0.9453	2568.64 390...	inst_129974/inst_495307
12	1.2000	0.9461	1.0244	0.9453	0.9453	2640.40 382...	inst_129974/inst_495150
13	1.2000	0.9461	0.9650	0.9451	0.9451	2607.28 384...	inst_129974/inst_491520
14	1.2000	0.9461	0.9541	0.9455	0.9455	2616.25 3810...	inst_129974/inst_491393
15	1.2000	0.9463	0.9936	0.9447	0.9447	2623.38 384...	inst_129974/inst_495108

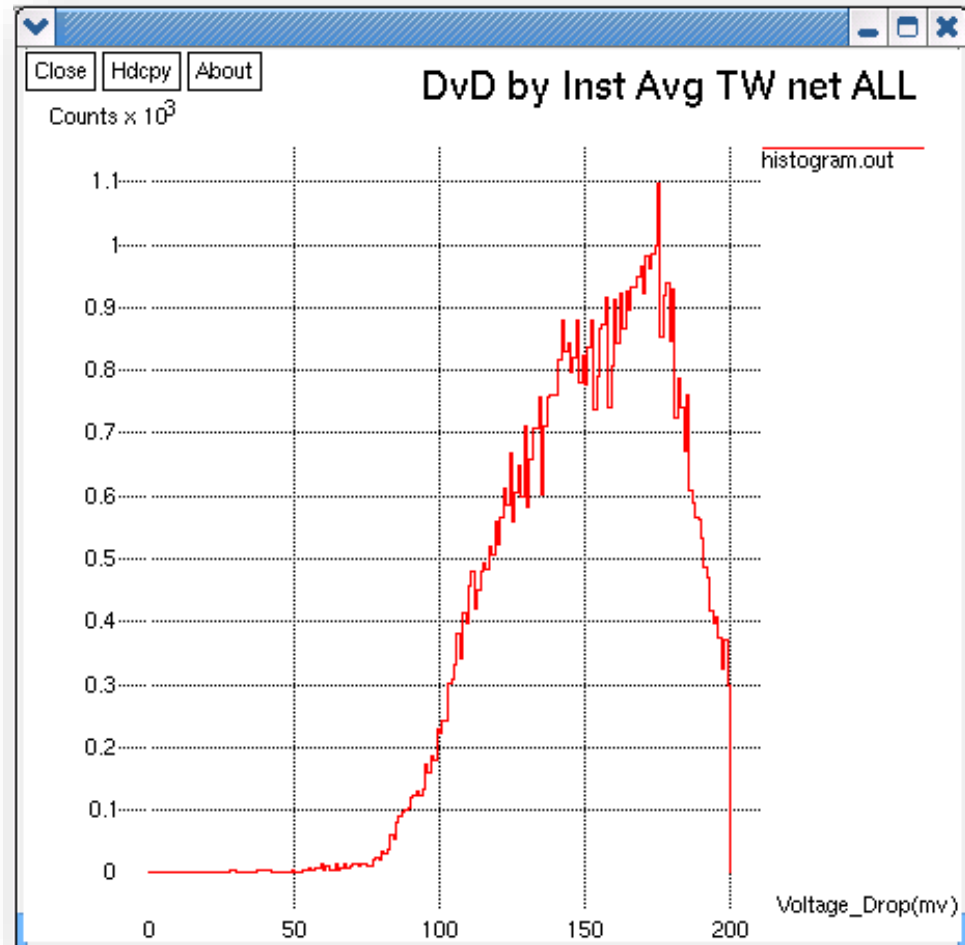
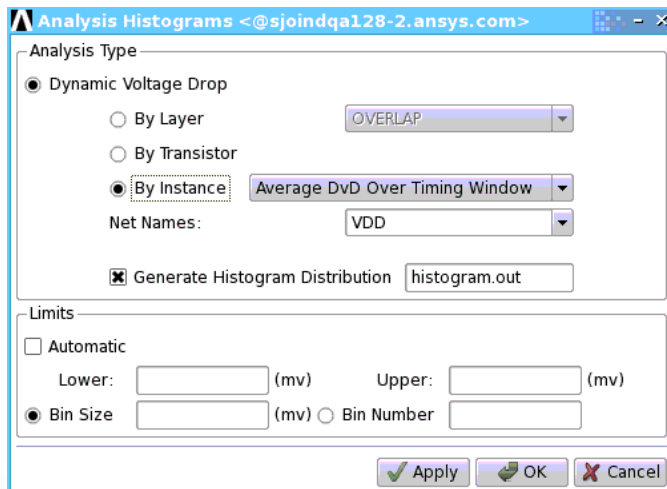
☒ Vdd Domain: All ☐ Ideal Vdd: 1.200 SortBy: Avg Vdd-Vss ☐ Threshold 5.00 %

Dynamic Voltage Drop Histogram

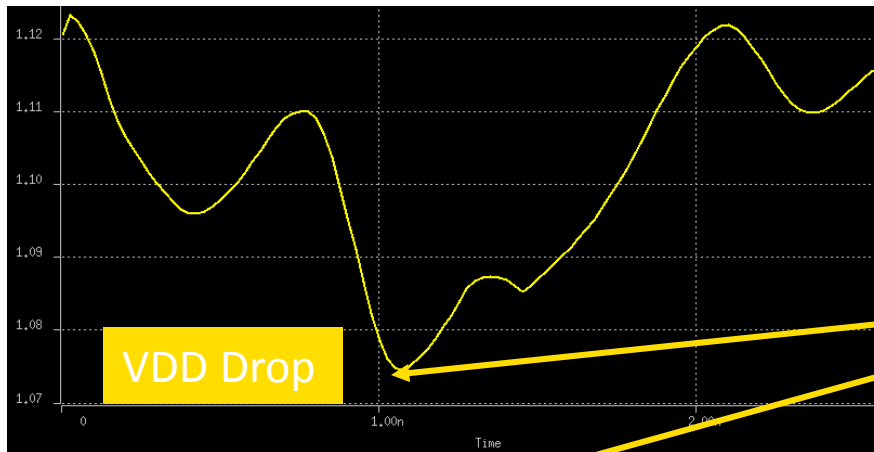
- TCL command for generating histogram

```
plot analysis -type DvD  
-instance avgTW  
-lower 0 -upper 200  
-binnumber 200
```

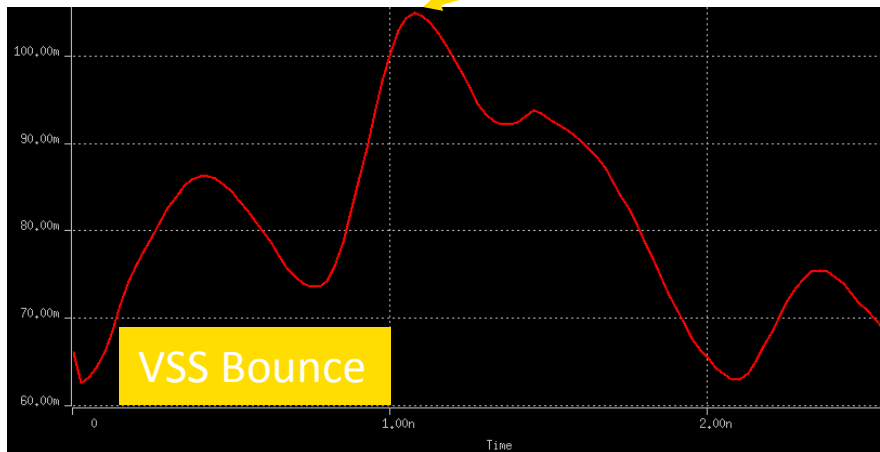
- GUI option
- Results - > Analysis histograms



Instance Voltage Waveform



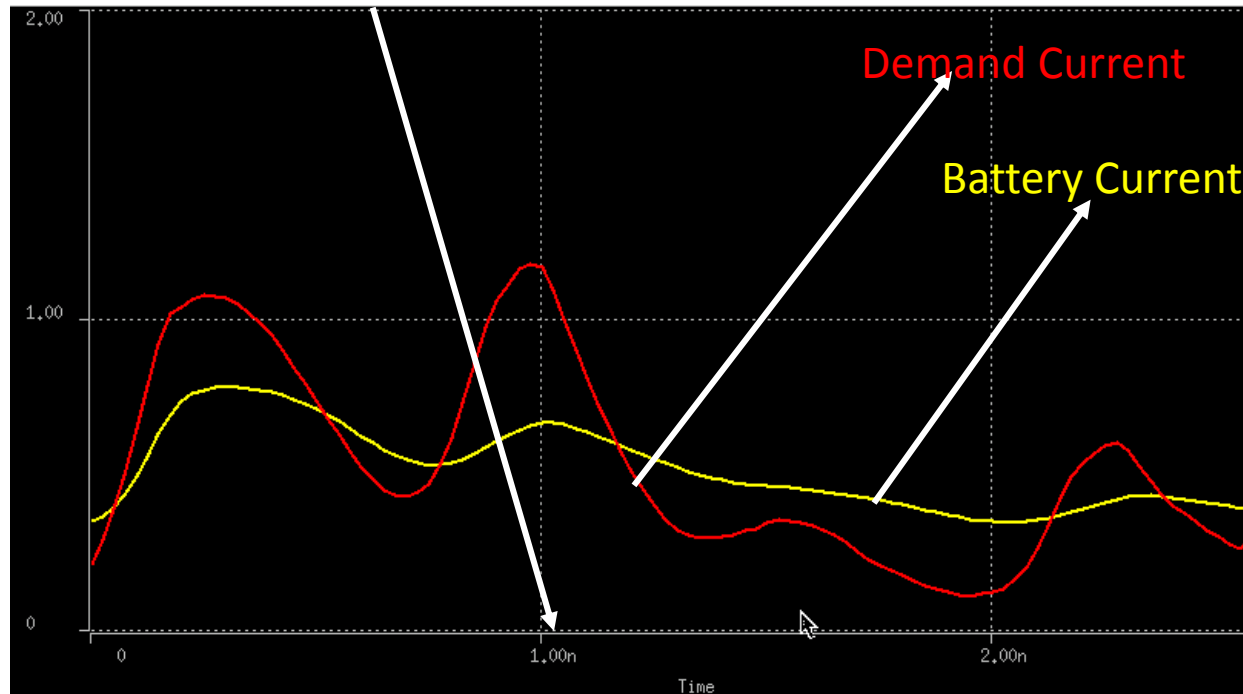
Max DvD at around 1ns !



```
plot voltage -name inst_129734 -sv
```

Current Waveforms

Max Current at around 1ns !

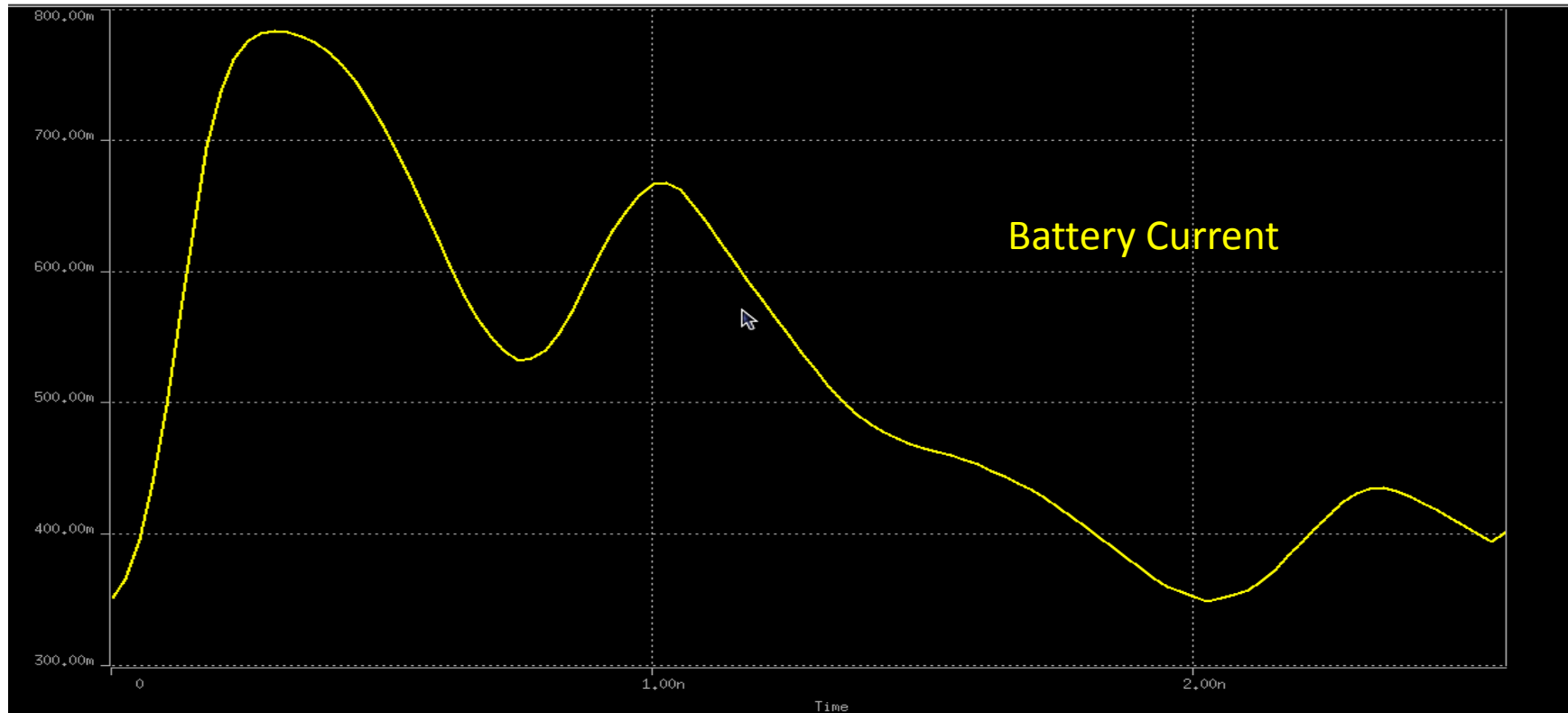


```
plot current -net -power -pad -overlay -sv
```

NOTE: Big difference between demand current and battery current

Indicates effectiveness of decaps

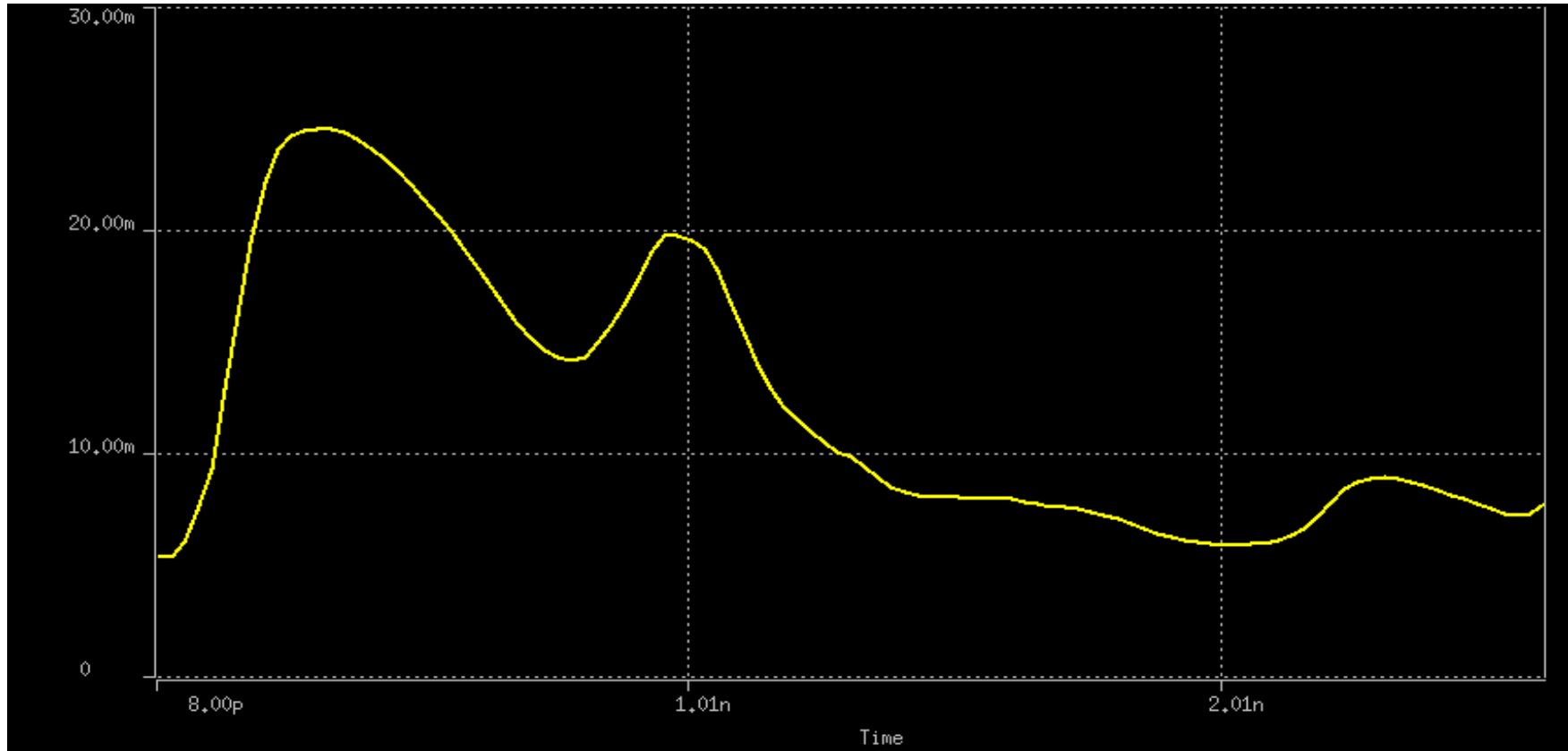
Current Waveforms (Cont'd)



```
plot current -net -name VDD -pad -sv
```

Current through all the pads with respect to net VDD

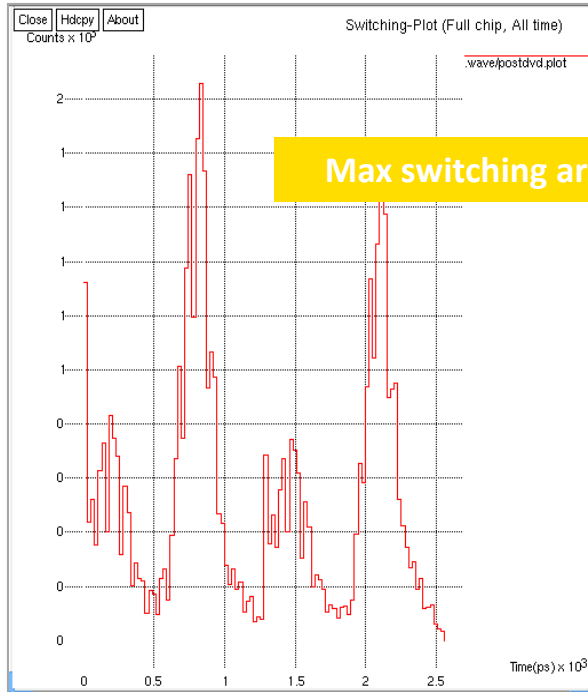
Current Waveforms (Cont'd)



```
plot current -pad -name DVDD1 -sv
```

Current through the pad DVDD1

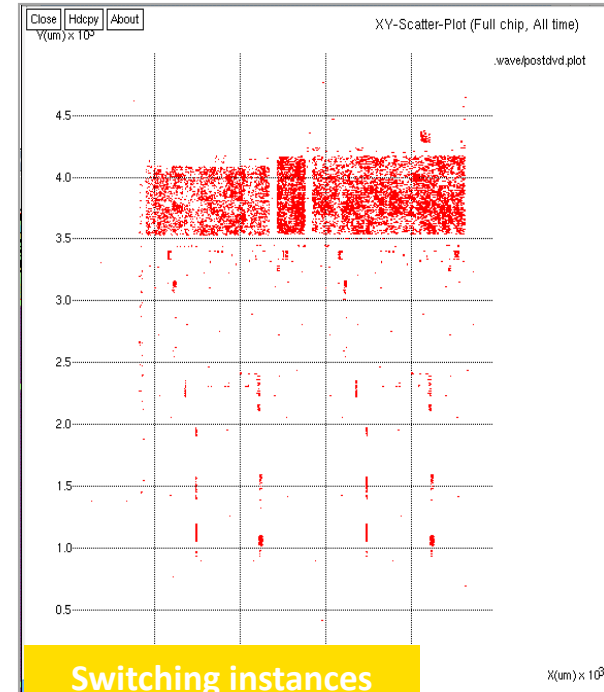
Switching Histogram



plot switching



plot charge



plot scatter

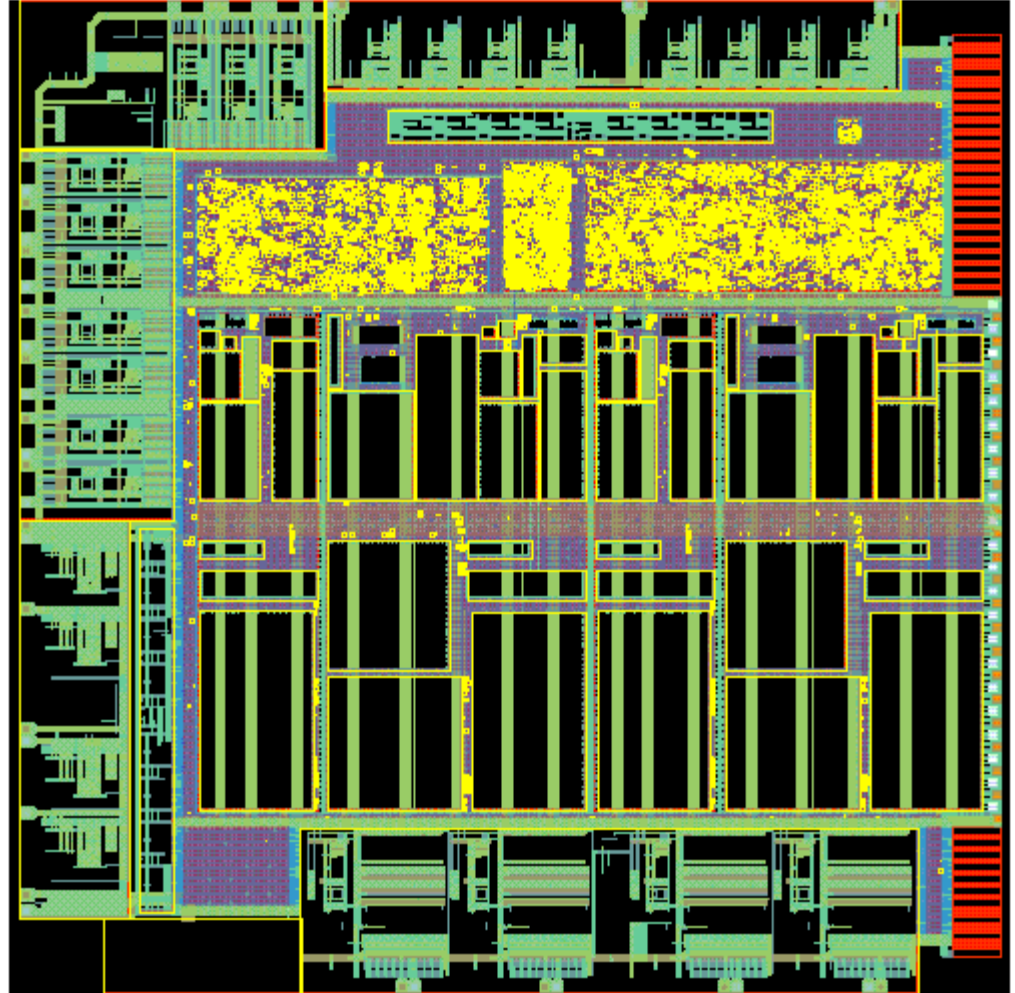
- Histograms can be generated for user given regions and time ranges with “condition set” command
- Plot switching w.rt specific cell_type can be obtained. First use "condition set -type [all | combinational | sequential | clock | memory]" to pick the cell type and then do a “plot switching”

Analyzing the Switching Events

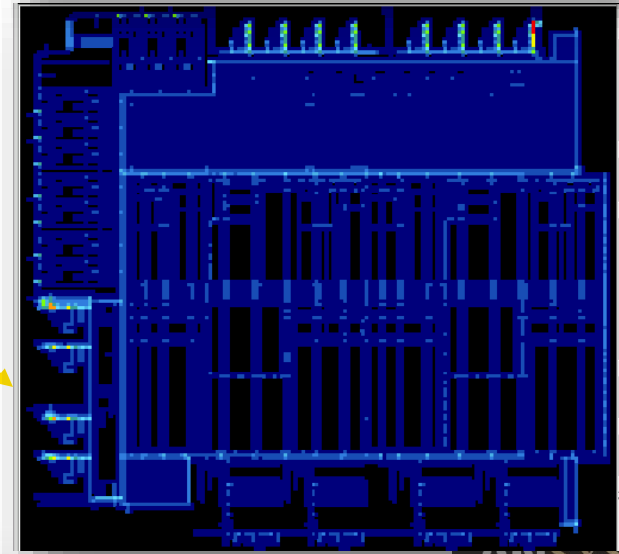
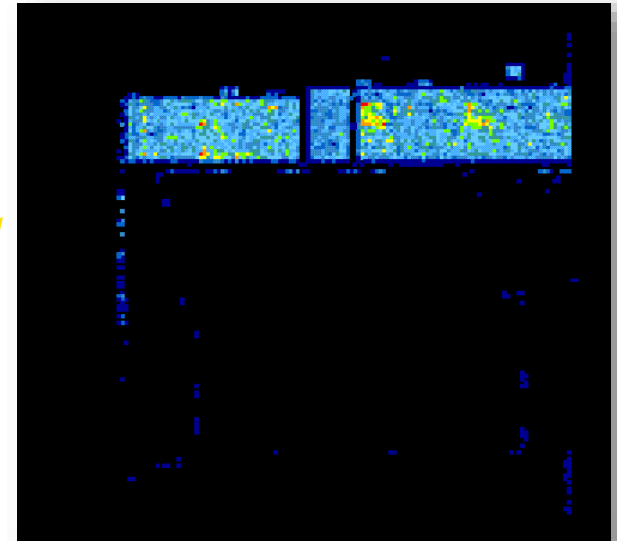
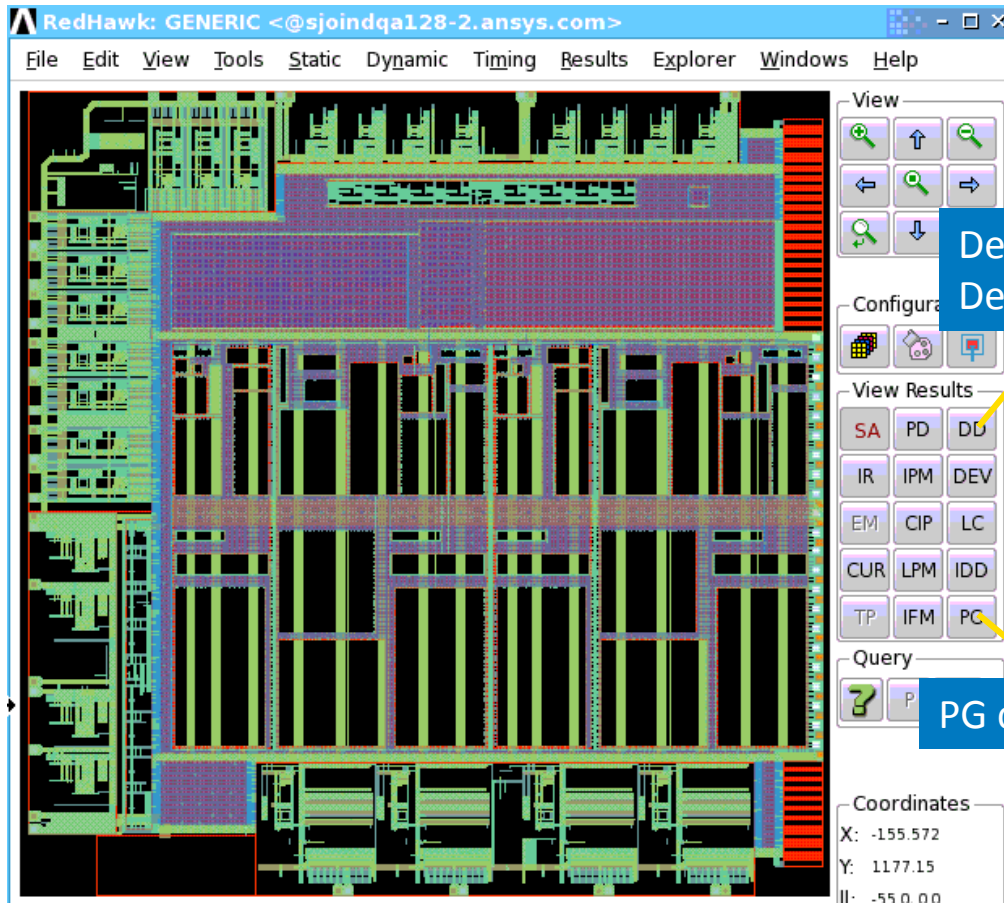
```
select add [print sw]
```

Command “print sw” lists all switching instances

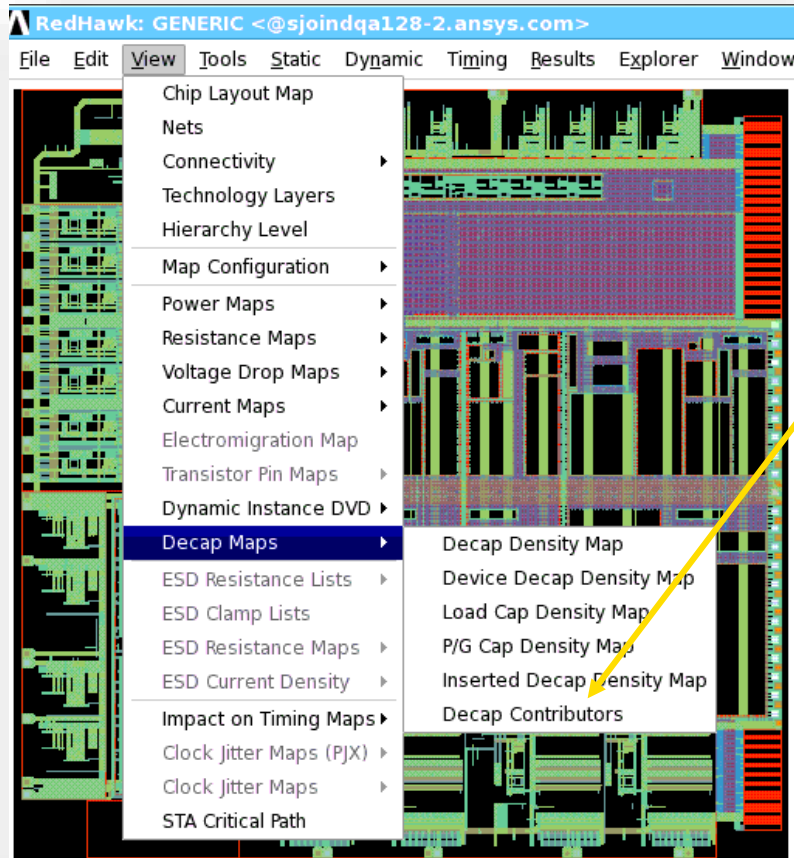
Simultaneous switching causes high voltage drop at around 1ns !!!



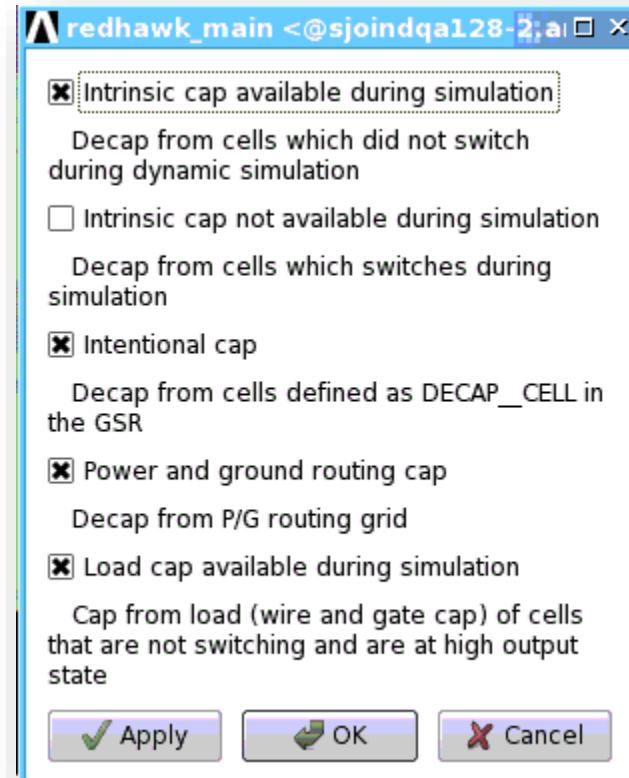
Decap Density Maps



Decap Density Maps (Cont'd)



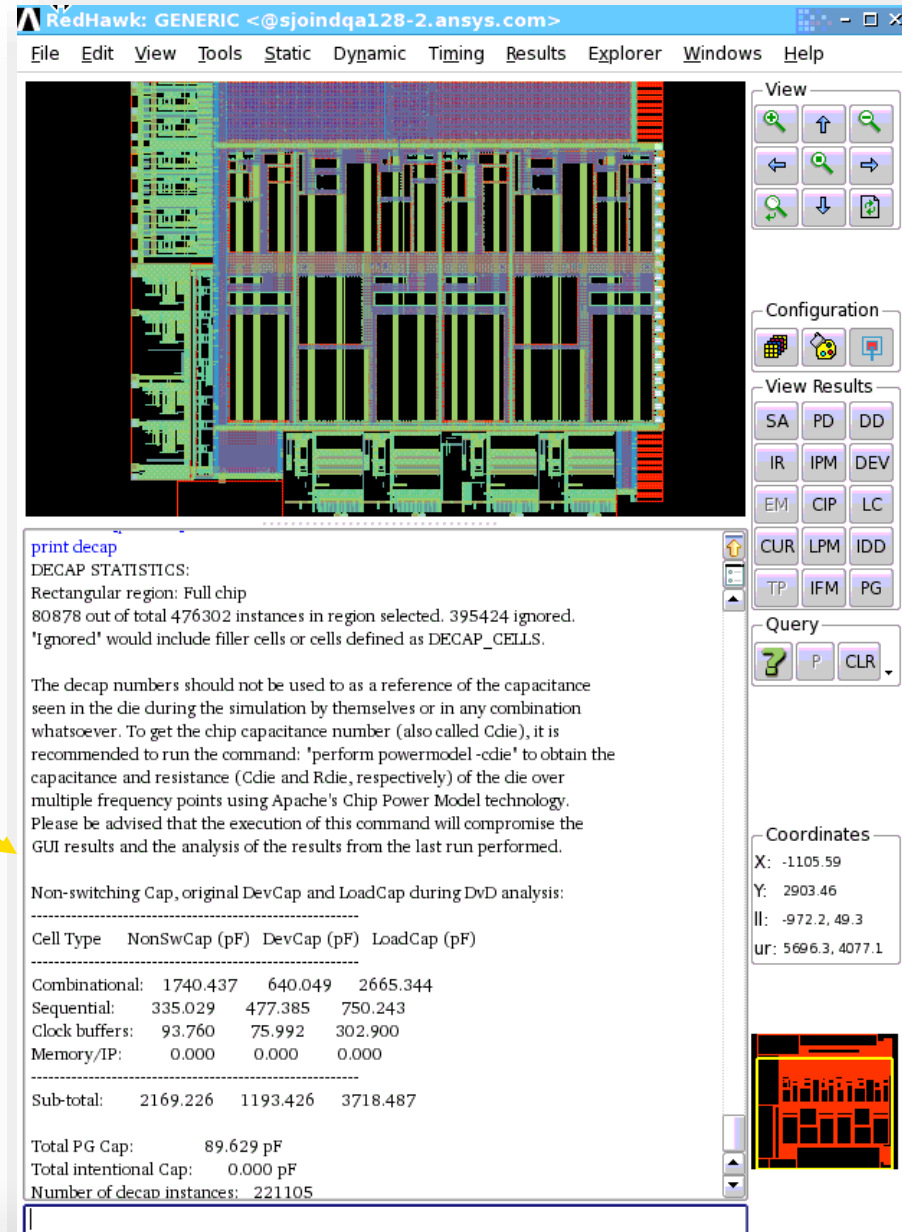
Select the decap contributors for Decap Density Map



Set GSR keyword DECAP_TILE_MAX for setting uniform tile size to all decap related maps

Decap Report

"print decap" command
showing detailed decap report



RedHawk: GENERIC <@sjoindqa128-2.ansys.com>

File Edit View Tools Static Dynamic Timing Results Explorer Windows Help

View

Configuration

View Results

SA PD DD
IR IPM DEV
EM CIP LC
CUR LPM IDD
TP IFM PG

Query

? P CLR

Coordinates

X: -1105.59
Y: 2903.46
ll: -972.2, 49.3
ur: 5696.3, 4077.1

print decap

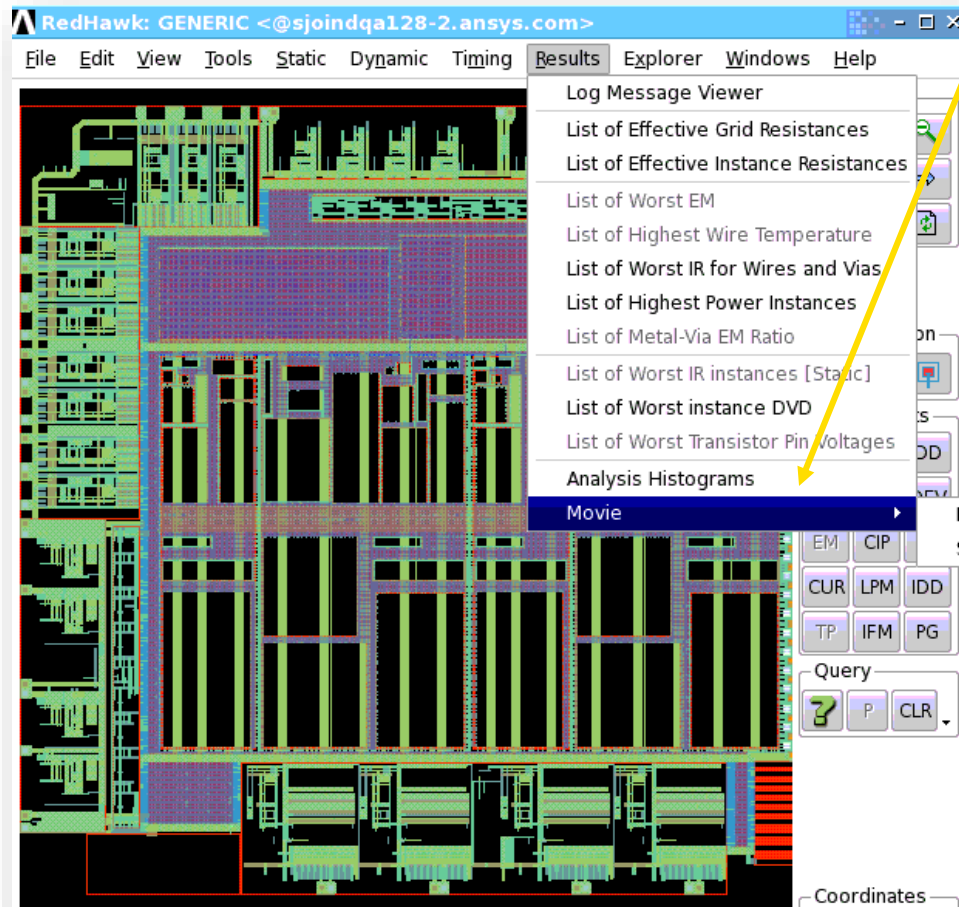
DECAP STATISTICS:
Rectangular region: Full chip
80878 out of total 476302 instances in region selected. 395424 ignored.
'Ignored' would include filler cells or cells defined as DECAP_CELLS.

The decap numbers should not be used as a reference of the capacitance seen in the die during the simulation by themselves or in any combination whatsoever. To get the chip capacitance number (also called Cdie), it is recommended to run the command: 'perform powermodel -cdie' to obtain the capacitance and resistance (Cdie and Rdie, respectively) of the die over multiple frequency points using Apache's Chip Power Model technology. Please be advised that the execution of this command will compromise the GUI results and the analysis of the results from the last run performed.

Non-switching Cap, original DevCap and LoadCap during DvD analysis:

Cell Type	NonSwCap (pF)	DevCap (pF)	LoadCap (pF)
Combinational:	1740.437	640.049	2665.344
Sequential:	335.029	477.385	750.243
Clock buffers:	93.760	75.992	302.900
Memory/IP:	0.000	0.000	0.000
Sub-total:	2169.226	1193.426	3718.487
Total PG Cap:		89.629 pF	
Total intentional Cap:		0.000 pF	
Number of decap instances:		221105	

Dynamic Voltage Drop Movie



➤ Dynamic Movie

➤ *Results -> Movie -> Make*



Dynamic Analysis Reports

adsRpt/Dynamic/GENERIC.dvd	Dynamic instance voltage drop
adsRpt/Dynamic/GENERIC.ir.worst	Wire voltage drop
adsRpt/Dynamic/pad.current	Current waveforms: Current through each pad
adsRpt/apache.refCell.noApl*	Lists of cells without APL data

Dynamic Analysis - Summary

- **Data requirements and setup are similar to static, except:**
 - needs switching time for instances
 - uses APL data for accuracy
- **True transient simulation**
- **Considers**
 - On-chip and off-chip capacitance and inductance
 - Simultaneous cell switching
- **Much more realistic modeling of voltage drop than static**

Result Exploration using RedHawk Explorer

Summary Section

RedHawk™-Explorer

ANSYS

SummaryData IntegrityDesign WeaknessHotspotsPerformance

BackForward3D-ICFAQ

Design Summary

Results Summary

Power Summary
Static Voltage Drop Summary
Dynamic Voltage Drop Summary
Current w/f Summary
Power EM Summary
Signal EM Summary
PathFinder Summary
Low Power Analysis Summary
CPM Summary

Run Details

Design Statistics

Key Parameters

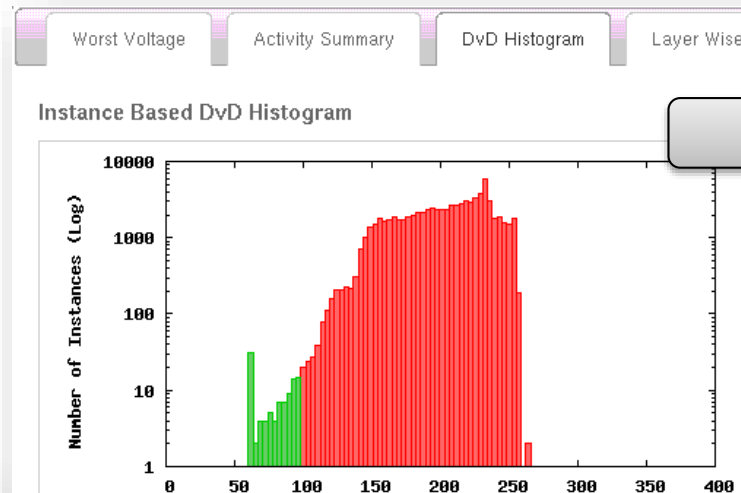
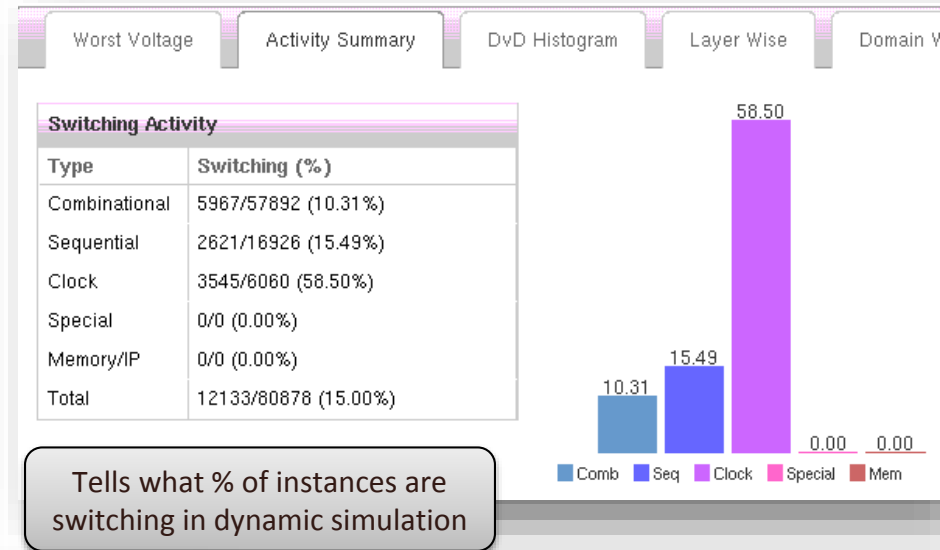
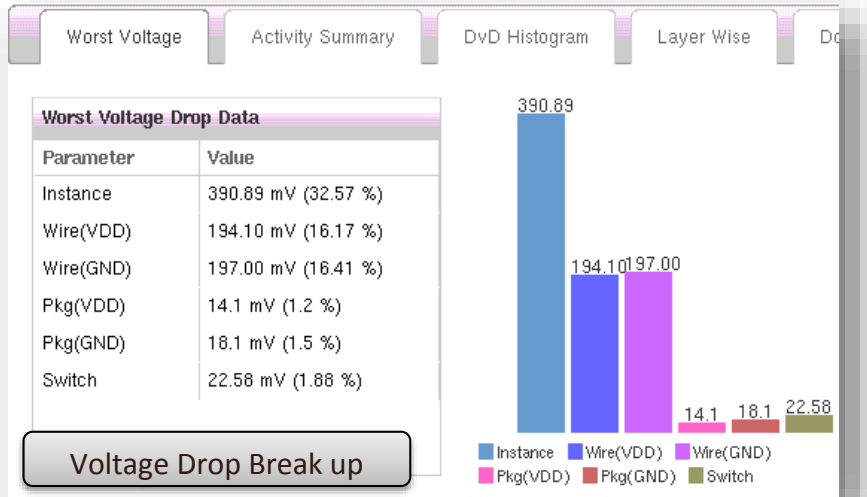
Design Name	GENERIC
Design Size	4920.62u x 5000.36u
Instance Count	476302
No of Voltage domains	2(Power) & 1(Ground)
Total Power	0.75 W

Design Summary
Result Summary
Power Summary
Dynamic Voltage Drop Summary
Power EM Summary
Current W/f Summary
Run Details

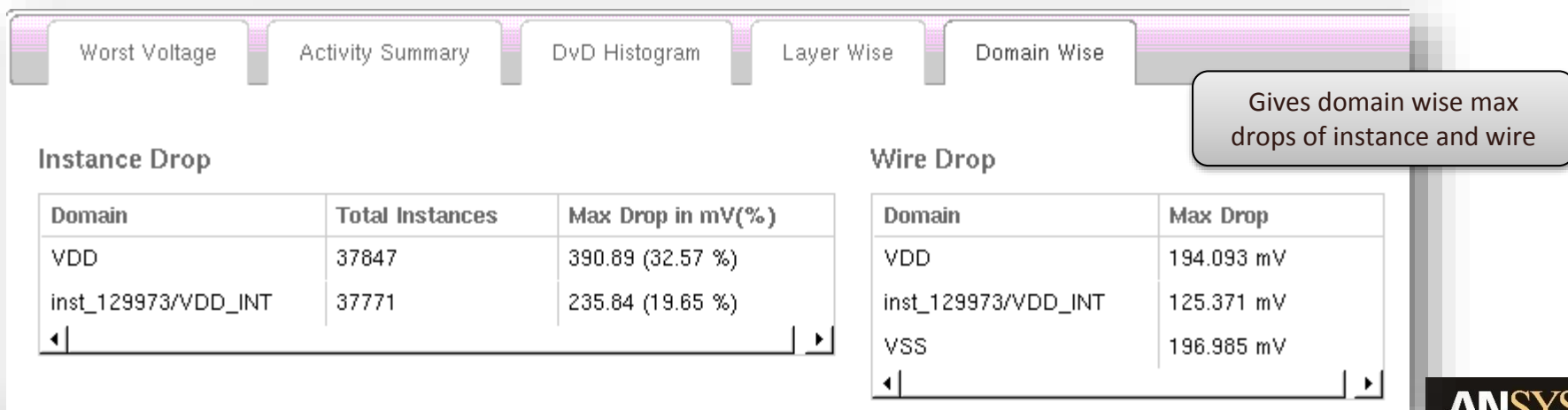
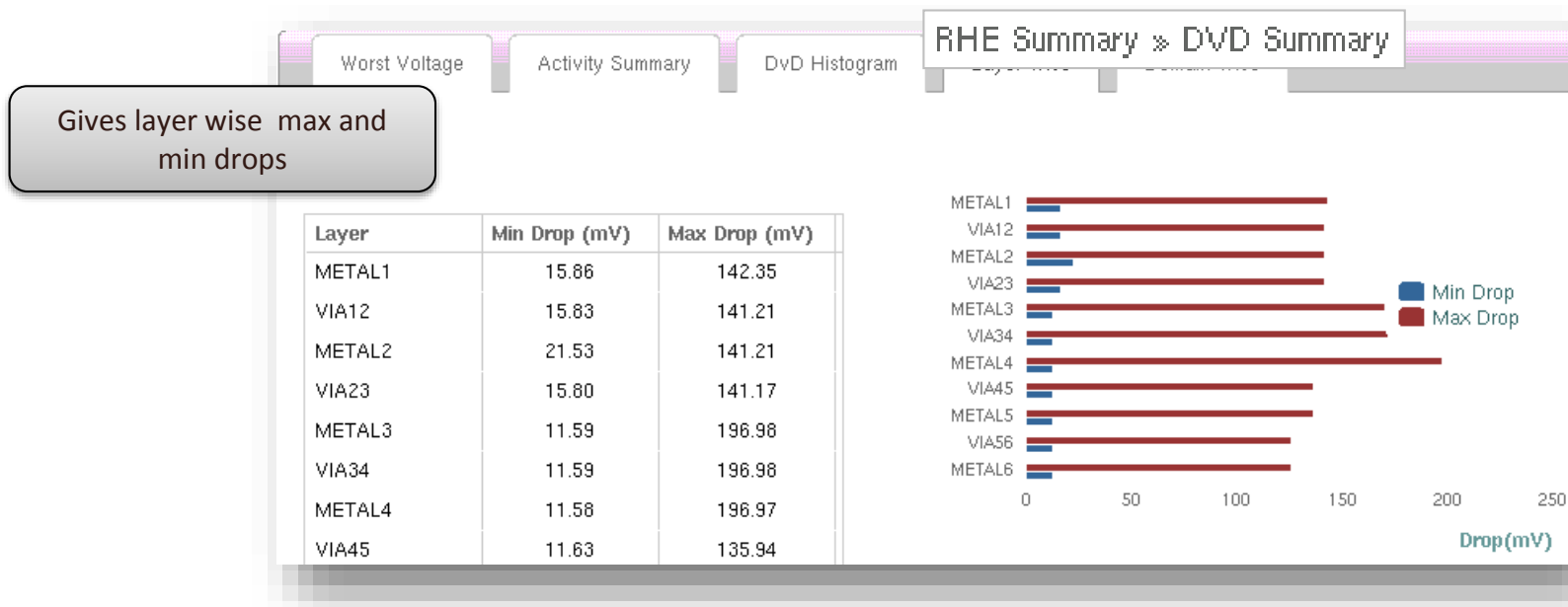
Chip Layout

DVD Summary

RHE Summary » DVD Summary



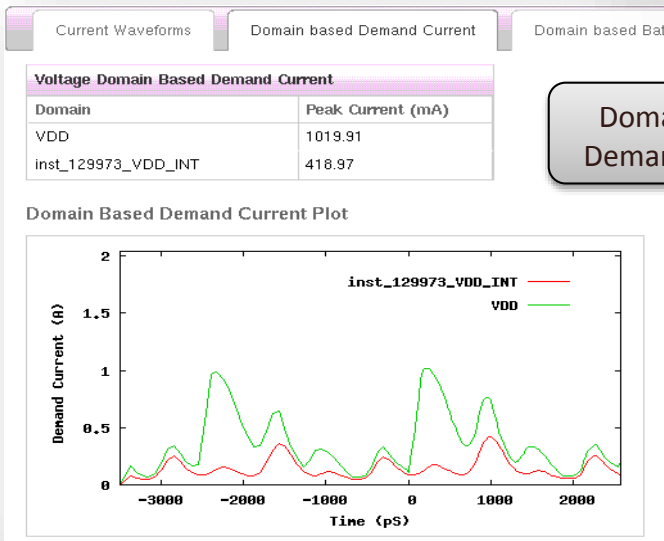
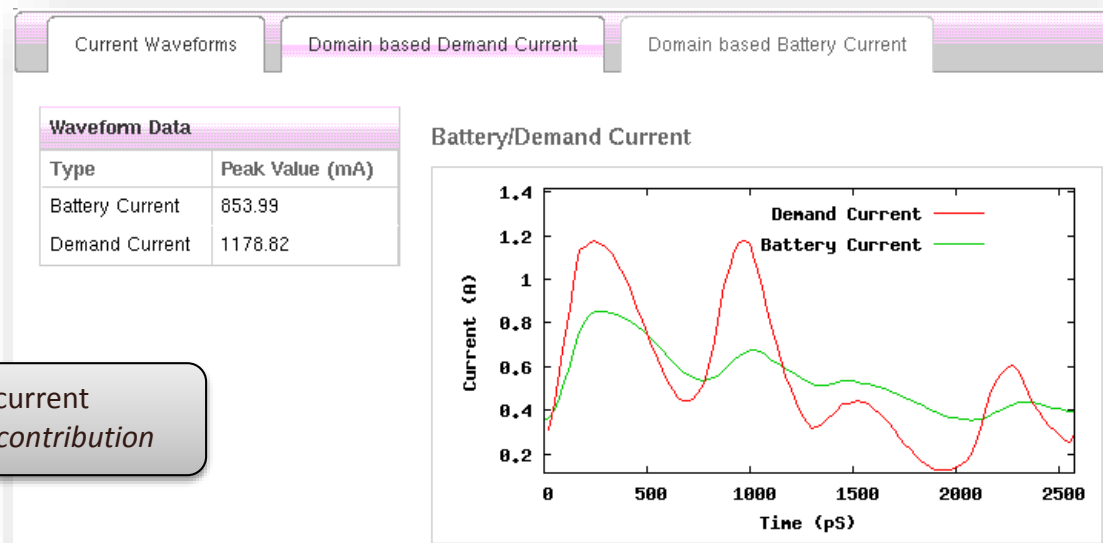
DVD Summary



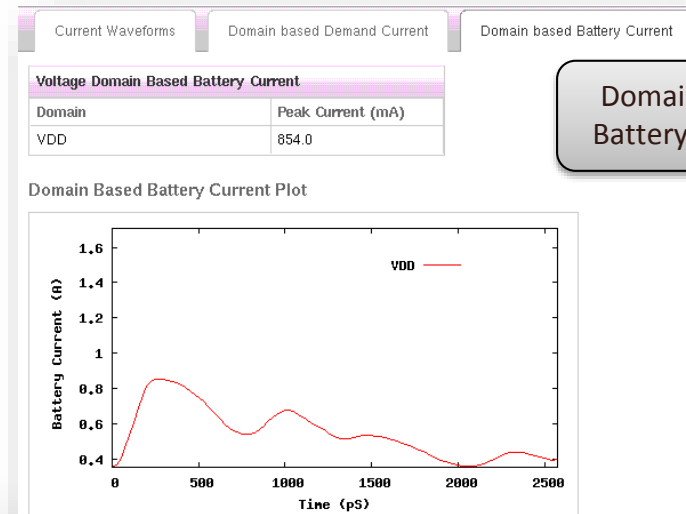
Current Waveforms Summary

RHE Summary » Current w/f Summary

Demand Current and Battery current
Difference indicates decap current contribution



Domain based
Demand Current



Domain based
Battery Current

Data Integrity Check Summary

Summary

Data Integrity

Design Weakness

Hotspots

Performance

Back

Forward

3D-IC

FAQ

Library Data

APL Current Check

✓

APL Cap Check

✗

APL Pwcap Check

✓

LIB Check

✓

LEF Check

✓

CMM Check

✓

Gds2def Check

✗

Design Data

DEF Check

✗

SPEF Check

✓

IPF Check

✓

STA Check

✓

VCD Check

✓

RedHawk Data

GSR Tcl Check

✗

Package Settings

✓

PAD_FILES

✓

GSC FILE

✓

Log Summary

ERROR Check

✗

WARNING Check

✗

Data Integrity Summary

Analysis Summary

Input Data	Status	Summary Description
☐ APL Current Check	✓	APL Current coverage is 99.08 %
☒ APL Cap Check	✗	APL Cap coverage is only 17.00 %
☒ SPEF Check	✓	SPEF coverage is 95.10 %
☒ STA Check	✓	STA coverage is 99.21 %
☒ LEF Check	✓	LEF coverage is 100.00 %
DEF Check	✗	DEF coverage is only 2.0 %
Package Check	✓	Package Subckt Is Present
GSR Tcl Check	✗	Details
Pad Files Check	✓	1) Number of P/G nets not driven by any p 2) Number of shorted pads : 0 3) Number of unused pads : 0 4) Number of floating pads : 0

RED cross indicates that there is a problem with this input

Violation Areas

Move the mouse pointer above the table rows to highlight the corresponding violation area.

- Design specific data integrity check
- Helps to identify and understand impact of missing data
- Breaks design into regions and presents missing data for each region

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41

Data Integrity Check : DEF Check

Missing Via Debug

Shorts Check Instance Unconnect Wire Unconnect

MISSING VIAS CHECK

- Check the number of missing vias in a design is not more than 10000

✓ Number of missing vias Reported : 116

Highlighting Regions with Missing Via in GUI

Zoom In RedHawk

Direct zoom into RH GUI by clicking on the image

RedHawk: GENERIC <@mercury>

File Edit View APL Tools Static Dynamic Timing Results Explorer He

Exact locations highlighted with marker

Highlights Regions with missing vias
(User can click and zoom into RH)

DEF Check: Related Reports

Data Integrity » Def Check » Related Reports

Def Check

Related Output Reports

Report	Line Count	Definition
adsRpt/apache.missingVias	116	Provides a list of missing via locations in the design
adsRpt/shorts.rpt	0	Provides a list of short locations in the design
adsRpt/GENERIC.power.unused	0	Provides list of instances that are not hooked to the power network
adsRpt/GENERIC.PG.unconnect	0	Provides list of instances belonging to analyzed PG network that are physically isolated from VDD and GND
		Provides list of in...

Data Integrity » Def Check » Related Reports » [adsRpt/apache.missingVias](#)

VSS METAL4 METAL3 805.23 2302.97 -0.00518688
VDD METAL4 METAL3 4242.55 2120.86 -0.00518692
VDD METAL4 METAL3 4078.84 2474.49 -0.00533938
VDD METAL4 METAL3 4077.41 2474.49 -0.00533938
VDD METAL4 METAL3 4075.97 2474.49 -0.00533938
VDD METAL4 METAL3 4817.96 1988.02 -0.00549197
VSS METAL4 METAL3 4820.38 2124.7 -0.00564455
VDD METAL4 METAL3 3961.59 3020.98 -0.00564456
VDD METAL4 METAL3 910.08 2263.74 -0.00579703
VSS METAL5 METAL4 4689.33 1991.86 -0.0057971
VDD METAL4 METAL3 3544.54 2484.56 -0.00579715

GSR/Tcl Check

Data Integrity » GSR TCL Check

GSR TCL Check

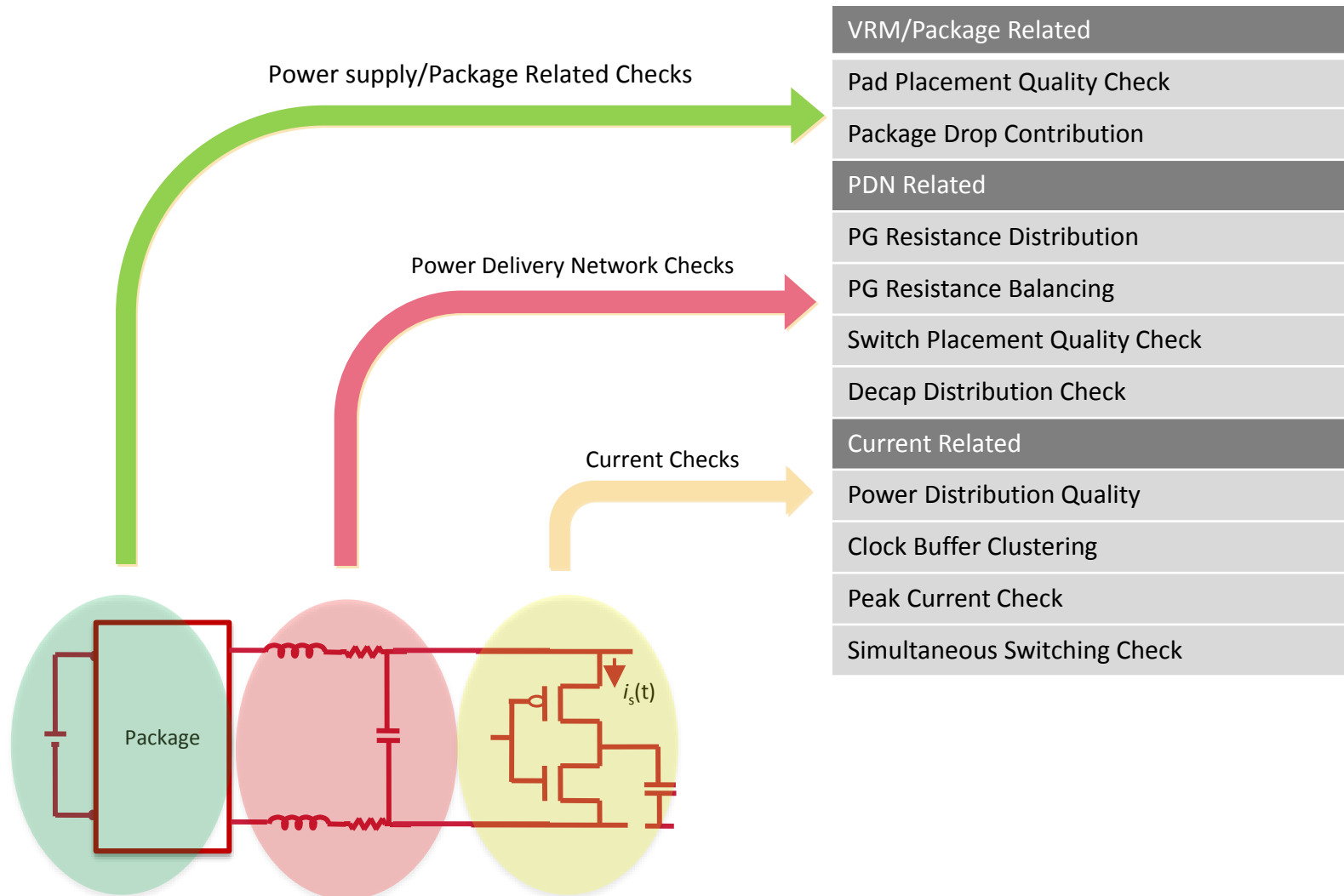
● Performs Checks on GSR/TCL Parameters

GSR ::Parameter	Given Value/Total	Recommended Value/Total	Status
GSR::PUSH_PININST	1	When set to 1 this keyword produces more accurate LEF modeling of power/ground pins	✓
GSR::DYNAMIC_SIMULATION_TIME	2.575 ns	> 2.56 ns	✓
GSR::DYNAMIC_TIME_STEP	25 ps	5 ps - 100 ps	✓
GSR::DYNAMIC_PRESIM_TIME	3 ns	1 ns - 50 ns	✓
GSR::FREQUENCY	390.0 MHz	~ 390.62 MHz	✓
GSR::CACHE_MODE	0	Set it to 1 if memory usage is a concern	🔥
GSR::DYNAMIC_SAVE_WAVEFORM	1	Set it to 0 if disk usage is a concern	🔥
GSR: INPUT_TRANSITION	3e-10	<= 1ns	✓
GSR : SPLIT_SPARSE_VIA_ARRAY	-1.0	.	✓
GSR::TOGGLE_RATE	AVG_CLOCK_TOGGLE_RATE : 0.64 AVG_SIGNAL_TOGGLE_RATE : 0.12	1.0 - 2.0 0.15 - 0.30	🔥 🔥
GSR::VDD_NETS	Total 2 power nets specified in GSR	There are 0 power nets missing in GSR	✓
GSR::VSS_NETS	Total 1 ground nets specified in GSR	There are 0 ground nets missing in GSR	✓

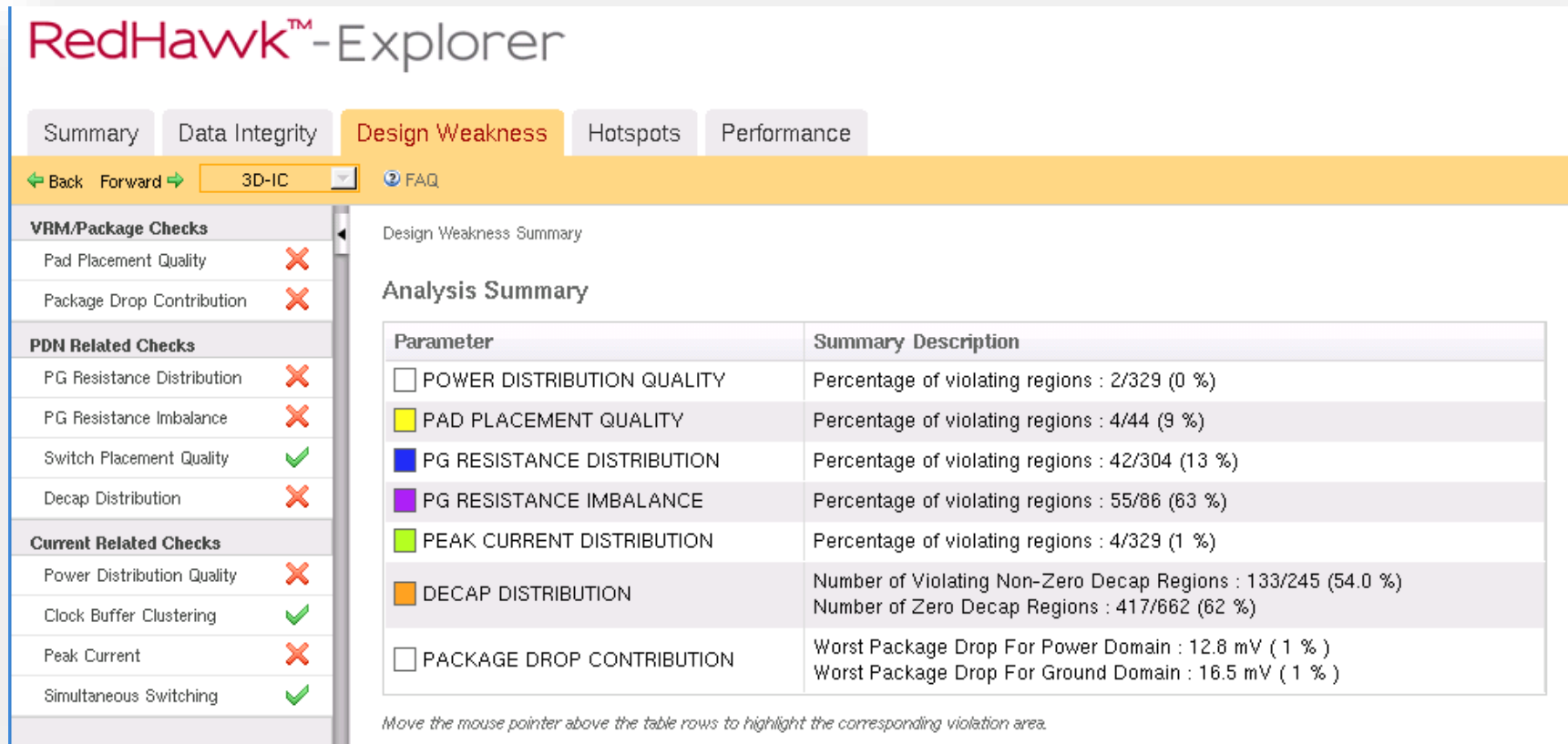
NOTE : GSR/TCL check is just an advisory section. There can be several cases where user wants to override the recommended settings displayed above; Purpose of this check is to identify gross issues made by the user like INPUT_TRANSITION keyword set to 100 instead of 100e-12 so that RedHawk interprets the default slew as 100 secs instead of 100ps

Common mistakes made in GSR/TCL are identified here

Overall Design Weakness Exploration



Design Weakness Checks Summary



The screenshot shows the RedHawk Explorer interface with the 'Design Weakness' tab selected. The left sidebar lists various checks under three categories: VRM/Package Checks, PDN Related Checks, and Current Related Checks. The main area displays an 'Analysis Summary' table with parameters and their corresponding violation percentages and counts. A note at the bottom of the table instructs the user to move the mouse pointer above the table rows to highlight the corresponding violation area.

RedHawk™ - Explorer

Summary | Data Integrity | **Design Weakness** | Hotspots | Performance

Back Forward 3D-IC FAQ

VRM/Package Checks

- Pad Placement Quality ❌
- Package Drop Contribution ❌

PDN Related Checks

- PG Resistance Distribution ❌
- PG Resistance Imbalance ❌
- Switch Placement Quality ✅
- Decap Distribution ❌

Current Related Checks

- Power Distribution Quality ❌
- Clock Buffer Clustering ✅
- Peak Current ❌
- Simultaneous Switching ✅

Design Weakness Summary

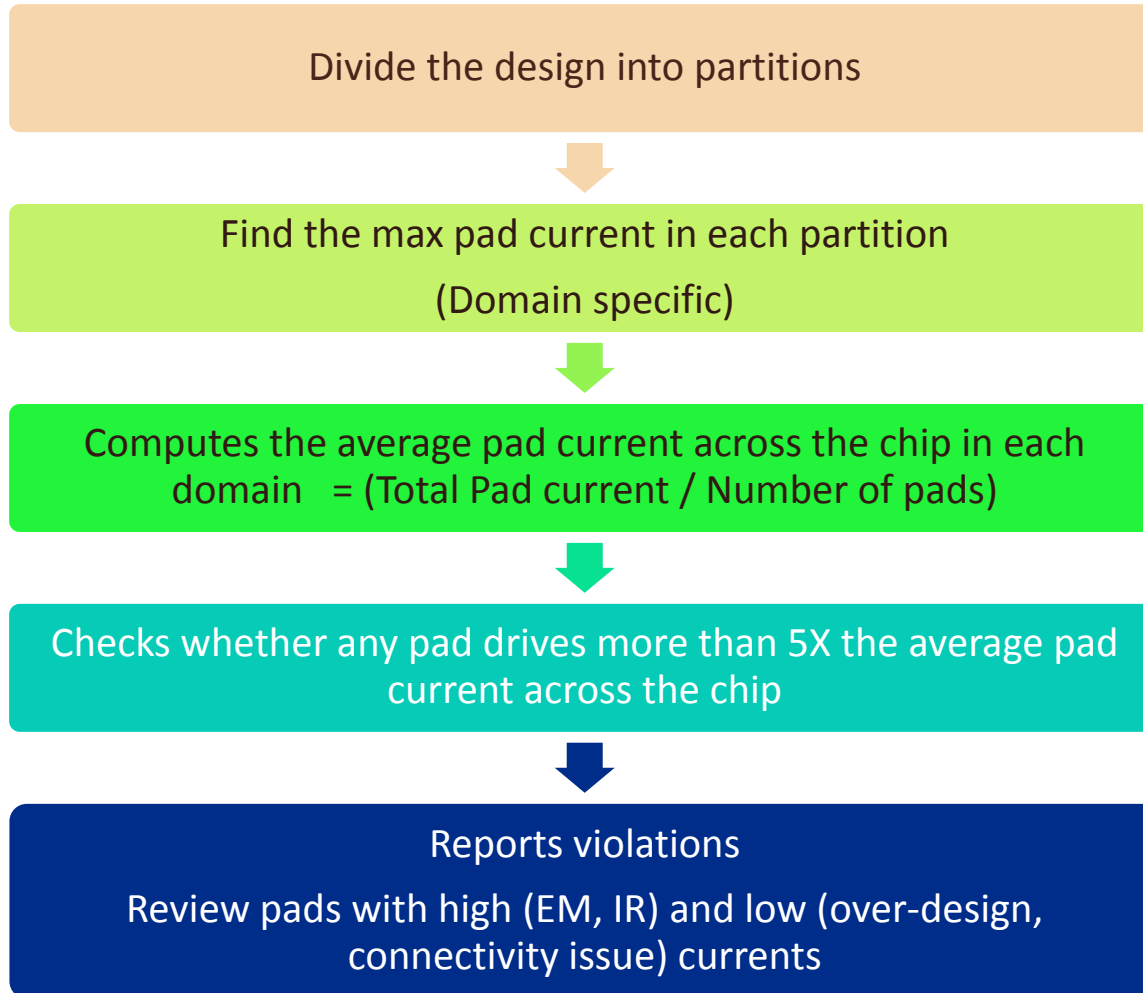
Analysis Summary

Parameter	Summary Description
☐ POWER DISTRIBUTION QUALITY	Percentage of violating regions : 2/329 (0 %)
☒ PAD PLACEMENT QUALITY	Percentage of violating regions : 4/44 (9 %)
☒ PG RESISTANCE DISTRIBUTION	Percentage of violating regions : 42/304 (13 %)
☒ PG RESISTANCE IMBALANCE	Percentage of violating regions : 55/86 (63 %)
☒ PEAK CURRENT DISTRIBUTION	Percentage of violating regions : 4/329 (1 %)
☒ DECAP DISTRIBUTION	Number of Violating Non-Zero Decap Regions : 133/245 (54.0 %) Number of Zero Decap Regions : 417/662 (62 %)
☐ PACKAGE DROP CONTRIBUTION	Worst Package Drop For Power Domain : 12.8 mV (1 %) Worst Package Drop For Ground Domain : 16.5 mV (1 %)

Move the mouse pointer above the table rows to highlight the corresponding violation area.

- Ranks Different Regions in the design for Various Design Weakness Parameters
- Identifies Regions Affected with Design Quality Issues

Pad Current Check Example



Pad Current Check

Summary

Data Integrity

Design Weakness

Hotspots

Performance

Back Forward 3D-IC FAQ

VRM/Package Checks

Pad Placement Quality

Package Drop Contribution

PDN Related Checks

PG Resistance Distribution

PG Resistance Imbalance

Switch Placement Quality

Decap Distribution

Current Related Checks

Power Distribution Quality

Design Weakness » Pad Placement Quality Check

Pad Placement Quality Check

Checks whether the Max Pad Current in any region is greater than 3 times Average Pad Current(28.100 mA)

Number of Regions with Violations : 4/44 (9.09 %)

Top Max Violations

Region	Pad Current Ratio	Details
1	4.87	Details
2	3.66	Details
3	3.55	Details

Issues Highlighted in GUI

RedHawk: GENERIC <@mercury> <2>

File Edit View APL Tools Static Dynamic Timing Results Explorer H

Details

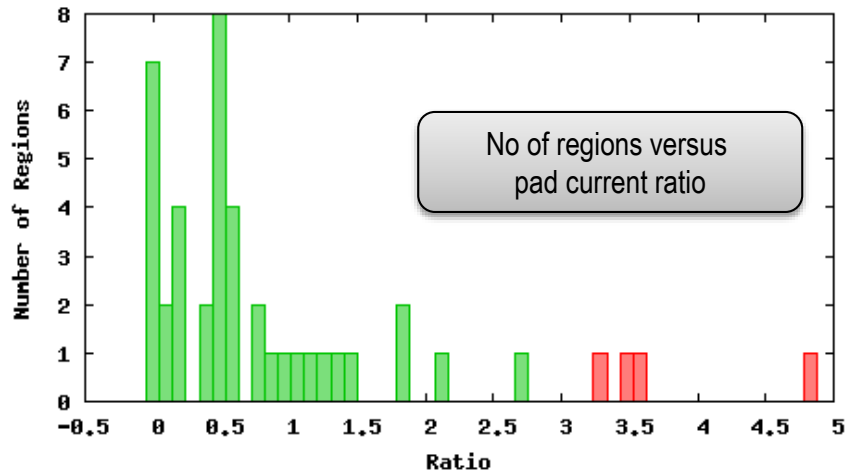
Parameter	Value
Region	4815.45 2324.70 4981.50 2490.75
Worst Pad	DVDD14
Worst Pad Location	4905 2438.85
Worst Pad Domain	VDD
Worst Pad Current	1.37e-01 A
Worst Pad Voltage	1.187185 V

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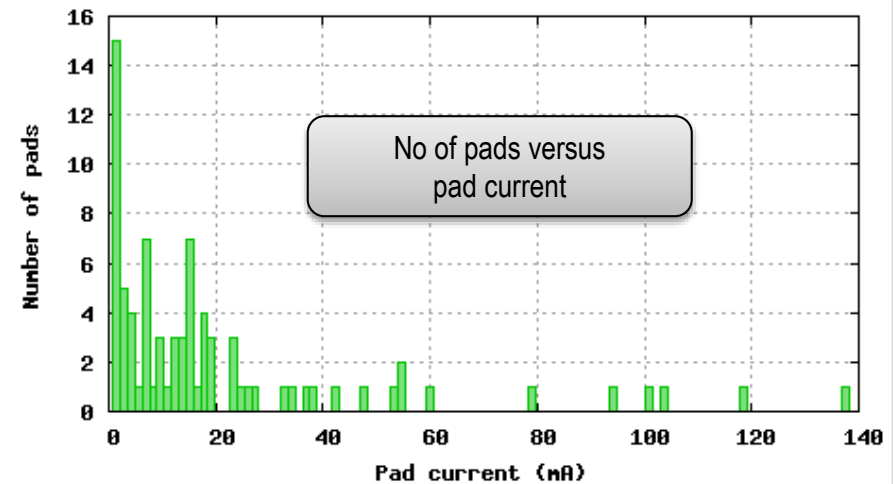
48

Pad Current Check - Details

Pad Current Distribution Histogram



Pad Current Histogram



Histogram

Voltage Drop

Domain Based Check

Cross Probing

VOLTAGE DOMAIN BASED PAD PLACEMENT QUALITY CHECK

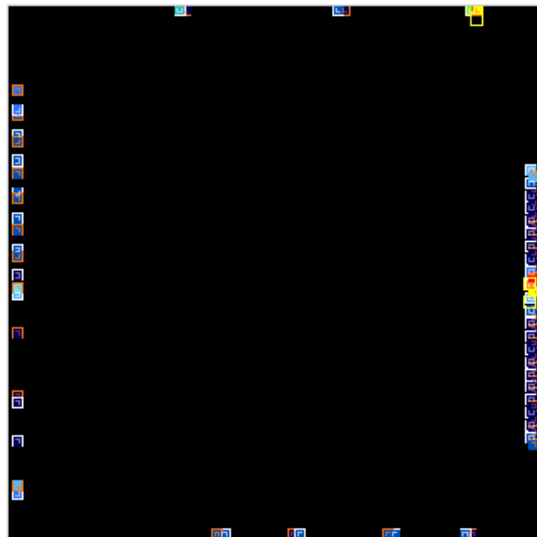
Domain	Avg Pad Current(A)	Max Pad Current(A)	Worst Pad
VDD	0.021129	0.136911	DVDD14
VSS	0.0217236	0.119063	DVSS13

Domain wise details

Pad Current Check - Details

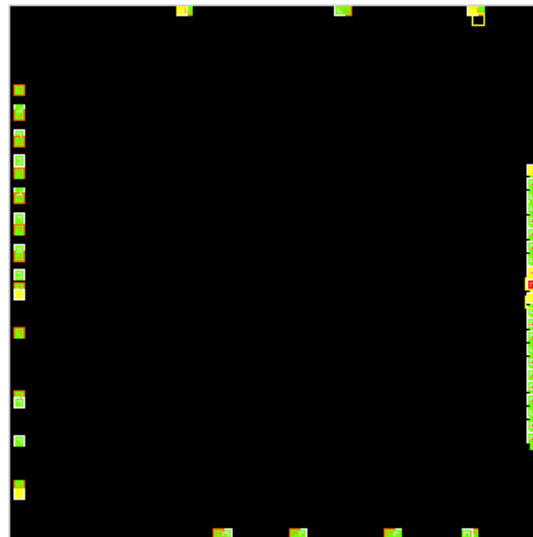
Design Weakness Summary » Pad Placement Quality Check » Related Maps

Design Weakness Summary » Pad Placement Quality Check » Related Maps



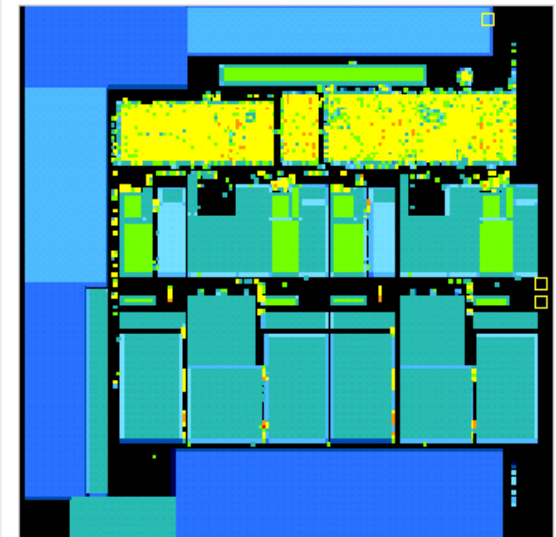
Pad Current Map

Show in RedHawk »



Pad Voltage Map

Show in RedHawk »



Power Density Map

Show in RedHawk »

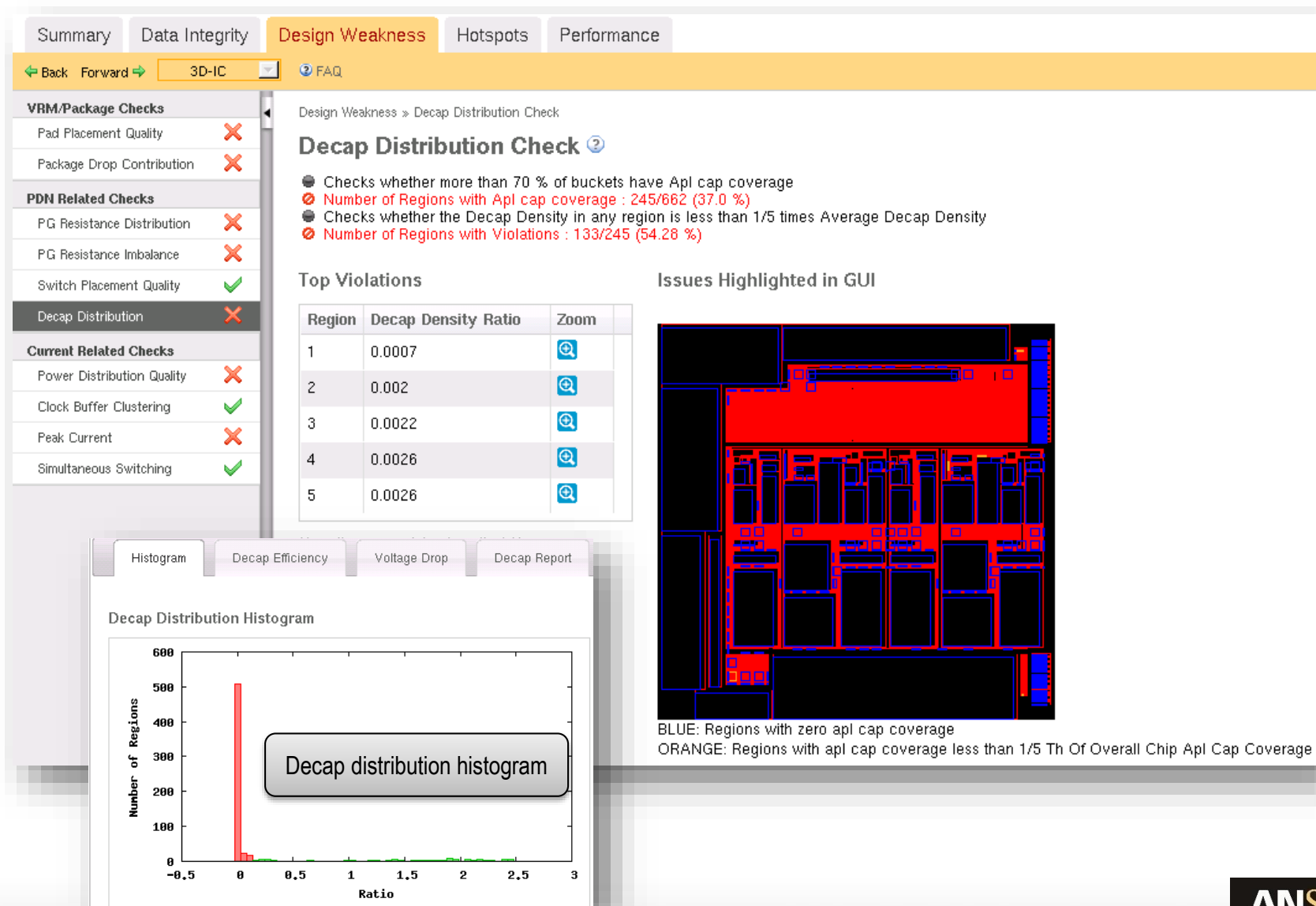
Data Integrity » Pad Placement Quality » Related Reports

Related Output Reports

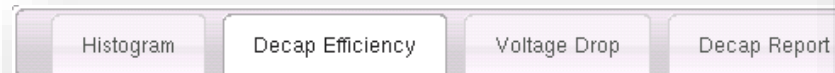
Report	Line Count	Definition
adsRHE/adsDWE/reports/pad.report	80	Provides details of current and voltage of each pad

Click on the hyperlink to
view the report

Decap Distribution Check

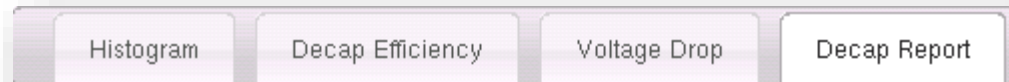
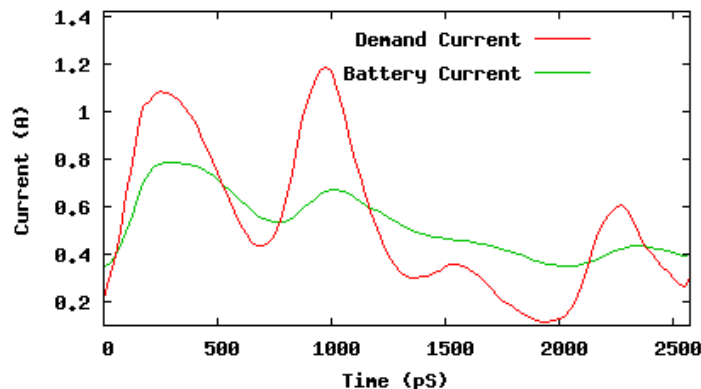


Decap Efficiency Check



DECAP EFFICIENCY CHECK

- Check whether the peak demand current is at least 20% more than the peak battery current is supplied by decaps)



DECAP STATISTICS:

Rectangular region: Full chip

80878 out of total 476302 instances in region selected. 395424 ignored.

"Ignored" would include filler cells or cells defined as DECAP_CELLS.

The decap numbers should not be used to as a reference of the capacitance seen in the die during the simulation by themselves or in any combination whatsoever. To get the chip capacitance number (also called Cdie), it is recommended to run the command: "perform powermodel -cdie" to obtain the capacitance and resistance (Cdie and Rdie, respectively) of the die over multiple frequency points using Apache's Chip Power Model technology. Please be advised that the execution of this command will compromise the GUI results and the analysis of the results from the last run performed.

Non-switching Cap, original DevCap and LoadCap during DvD analysis:

Cell Type	NonSwCap (pF)	DevCap (pF)	LoadCap (pF)
-----------	---------------	-------------	--------------

Combinational:	1743.752	640.049	2666.374
Sequential:	337.494	477.385	748.005
Clock buffers:	92.140	75.992	302.900
Memory/IP:	0.000	0.000	0.000

Sub-total:	2173.386	1193.426	3717.279
------------	----------	----------	----------

Total PG Cap: 89.632 pF
 Total intentional Cap: 0.000 pF
 Number of decap instances: 221105

Simultaneous Switching Check

Summary

Data Integrity

Design Weakness

Hotspots

Performance

← Back Forward → 3D-IC ? FAQ

VRM/Package Checks

Pad Placement Quality ✗

Package Drop Contribution ✗

PDN Related Checks

PG Resistance Distribution ✗

PG Resistance Imbalance ✗

Switch Placement Quality ✓

Decap Distribution ✗

Current Related Checks

Power Distribution Quality ✗

Clock Buffer Clustering ✓

Peak Current ✗

Simultaneous Switching ✓

Design Weakness » Simultaneous Switching Check

Simultaneous Switching Check ?

Switching Activity

Demand Current Check

Switching Event Clustering

● OVERALL SWITCHING ACTIVITY CHECK

● Checks If Overall Number Of Switching Instances Is Between 10 and 30 % Of Total Number Of Instances

Switching Report

Type	Switching/Total	Switching %
Combinational	6006/57892	10.37%
Sequential	2729/16926	16.12%
Clock	3553/6060	58.63%
Special	0/0	0.00%
Memory/IP	0/0	0.00%
Total	12288/80878	15.19%

Switching Activity Details

✓ Number Of Instances Switching : 12288/80878 (15.19 %)

● Celltype Based Switching Check

● Frequency Domain Based Demand Current Check

● Voltage Domain Based Demand Current Check

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53

Simultaneous Switching Check

Simultaneous Switching Check ②

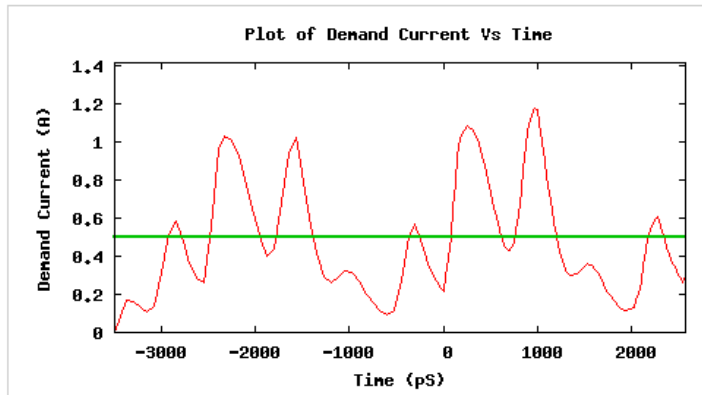
Switching Activity

Demand Current Check

Switching Event Clustering

- DEMAND CURRENT CHECK
- Check whether Peak Demand Current is greater than 10 times the Average demand current
- ✓ Chip Peak current to Average current ratio = 2.34 (1181.5 mAmp / 502.86 mAmp)

Demand Current Waveform



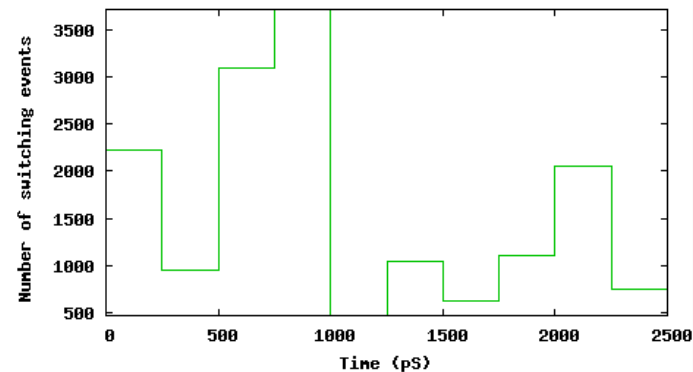
Demand Current Check

Switching Activity

Demand Current Check

Switching Event Clustering

- SWITCHING EVENT CLUSTERING CHECK
- Check whether Number of Switching events in a time-bin is greater than 5 X the Average Number of Switching Events per time-bin (1602.5)
- ✓ Number Of Failed Time-Bins 0/10 (0.0 %)



Switching Event Clustering
Check

Simultaneous Switching Check

Frequency Domain based Demand Current Check

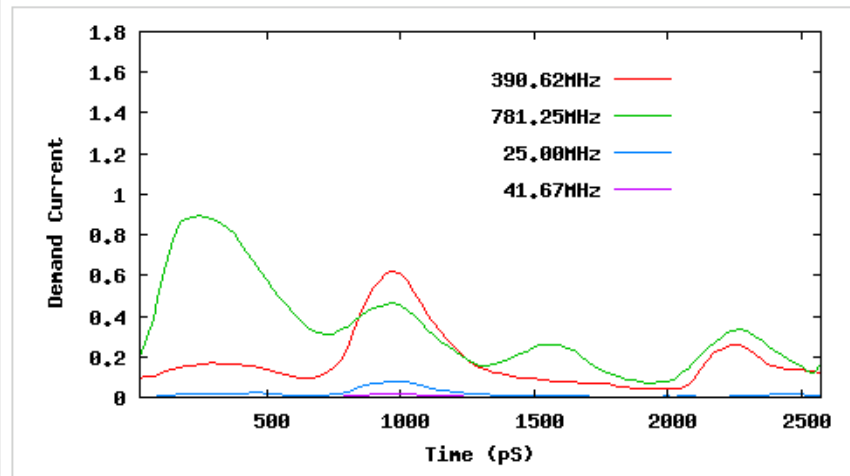
Design Weakness » Simultaneous Switching Check » Frequency Based

Simultaneous Switching Check ?

Frequency Domain Based Demand Current Check

Frequency	Peak Current	Ratio	Status
25.00MHz	80.6	4	✓
781.25MHz	899.94	2	✓
390.62MHz	621.91	3	✓
41.67MHz	19.33	5	✓

Frequency Domain Based Demand Current Plot

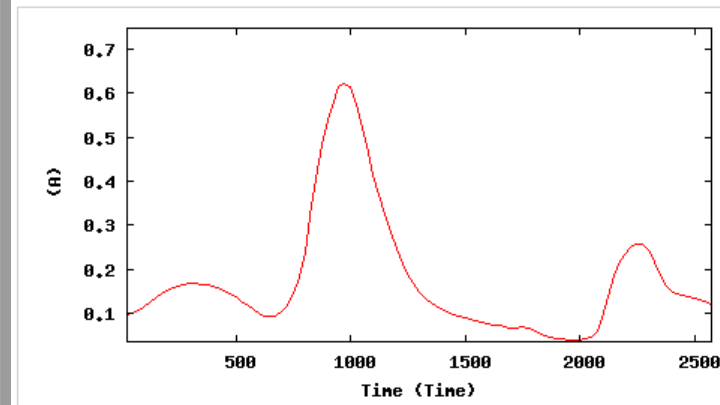


Design Weakness » Simultaneous Switching Check » Simultaneous Switching Check For Frequency inst_129973_VDD_INT

Simultaneous Switching Check ?

- Demand Current Check For freq : 390.62MHz
- Check whether Peak Demand Current for this frequency is greater than 10 times the Average demand current for this frequency
- ✓ Chip Peak current to Average current ratio = 3 (621.91 mA / 177.99 mA)

Demand Current Waveform For Frequency 390.62MHz



Simultaneous Switching Check

Voltage Domain Based Demand Current Check

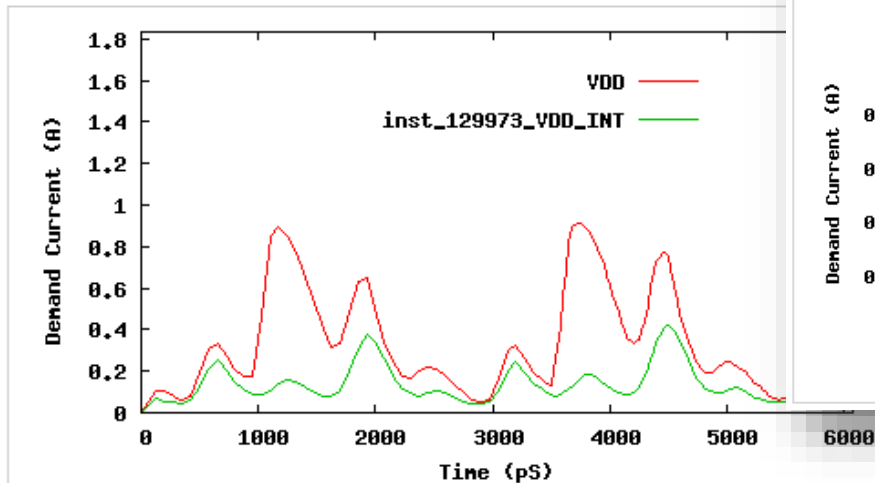
Design Weakness » Simultaneous Switching Check » Domain Based

Simultaneous Switching Check ?

Voltage Domain Based Demand Current Check

Domain	Peak Current (mA)	Ratio	Status
VDD	916.89	2	✓
inst_129973_VDD_INT	418.12	2	✓

Voltage Domain Based Demand Current Plot

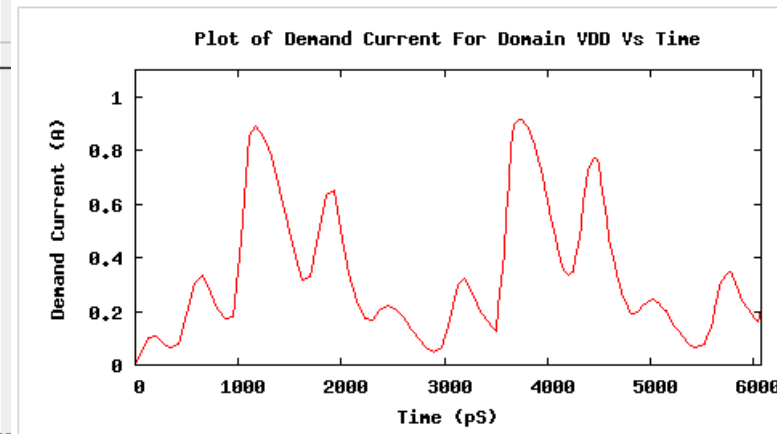


Design Weakness » Simultaneous Switching Check » Simultaneous Switching Check For Domain VDD

Simultaneous Switching Check ?

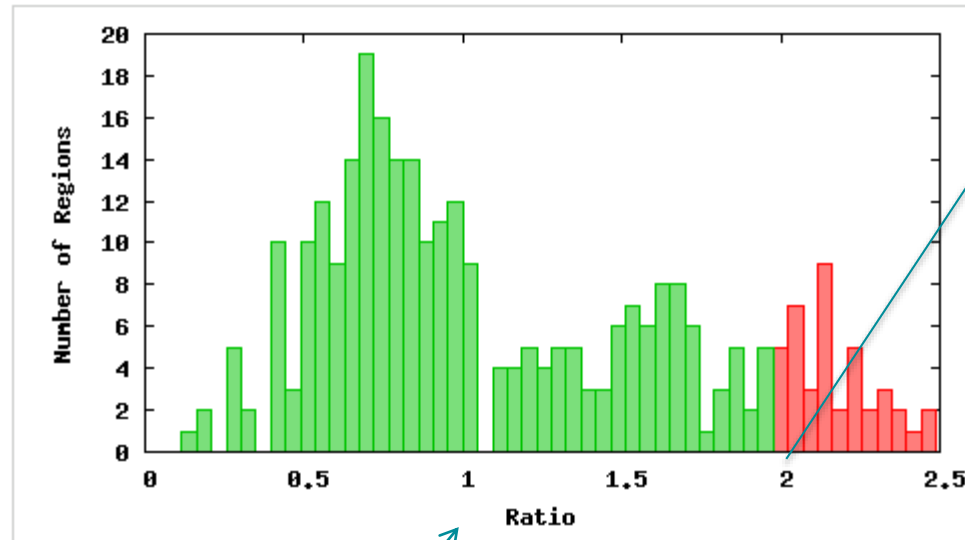
- Demand Current Check For Domain : VDD
- Check whether Peak Demand Current for this domain is greater than 10 times the Average demand current for this domain
- ✓ Chip Peak current to Average current ratio = 2 (916.89 mA / 355.0 mA)

Demand Current Waveform For Domain VDD



Interpreting Histograms in RHE

PG Resistance Distribution Histogram



Simple interpretation:
Ratio 2 means this region is 2X weaker
than other part of the design

How is this ratio computed in this example ?

- STEP-1 : RHE computes the peak resistance for each region (REGION-R)
- STEP-2 : Average resistance across all the regions is computed (AVG-R)
- STEP-3 : PGR Design Weakness Ratio for each region is computed as $\text{REGION-R}/\text{AVG-R}$

Cross-probing Violations in RedHawk GUI

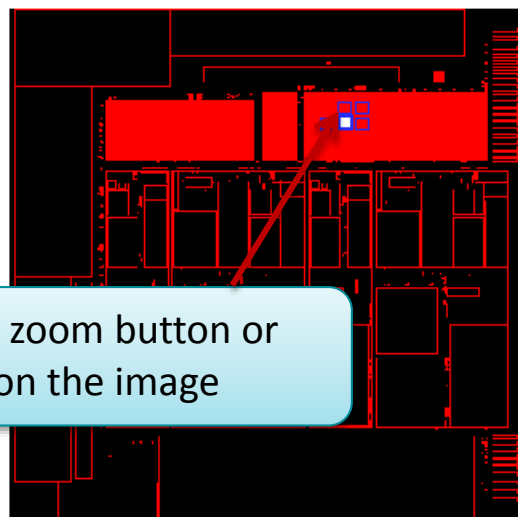
PG Resistance Distribution Check [?]

- Checks whether the PG Resistance in any region is greater than 2 times Average PG Resistance(24.350 ohm)
- ❌ Number of Regions with Violations : 42/304 (13.81 %)

Top Violations

Region	PG Resistance Re	Zoom	Details
1	2.94		Details
2	2.91		Details
3	2.86		Details
4	2.85		Details
5	2.84		Details

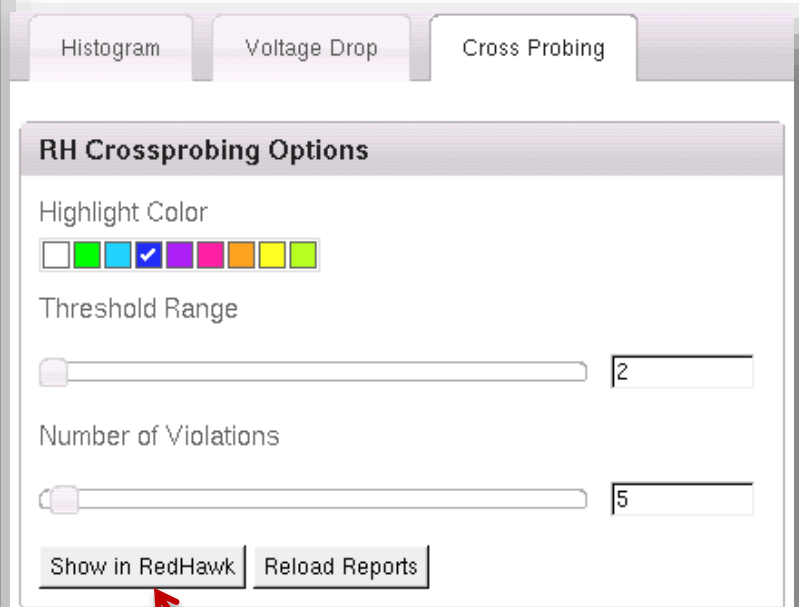
Issues Highlighted in GUI



User can click on zoom button or directly click on the image

When you click on the image, RH/RHE does 3 things:

- Zooms into the violating regions
- Highlights the worst instance within the regions
- Update the map with some relevant views
 - Example, PG resistance map will display SPT for the worst instance



Click on the Show in RedHawk button to view the violations in RH GUI

Hotspot Analysis Summary

Summary

Data Integrity

Design Weakness

Hotspots

Performance

Back

Forward

3D-IC

FAQ

Dynamic Voltage Drop Check

Static IR Check

Power EM Check

Signal EM Check

RMS

AVG

Peak

Max Cap Check

Memory Check

DVD Check

Static IR Check

Power EM Check

Low Power Check

Voltage & Current w/f

Differential Voltage Check

Noise Coupling Check

Switch Id-sat Check

Switch Off State Check

CPM

GSR/Tol Settings

Average Current Check

FFT Check

HotSpot Summary

Analysis Summary

Parameter	Summary Description
☐ Dynamic Voltage Drop Check	Check Not Performed
☒ Static IR CHECK	Worst Static Drop : 213.19 mV
☒ Power EM CHECK	Worst EM violation : 1523.20 %
☐ Low Power CHECK	Check Not Performed

Move the mouse pointer above the table rows to highlight the corresponding Hotspots

Violation Areas

DVD Check : Summary

RedHawk™-Explorer

Summary Data Integrity Design Weakness Hotspots Performance

Back Forward 3D-IC FAQ

Dynamic Voltage Drop Check ✖

Static IR Check

Power EM Check ✖

Signal EM Check

RMS

AVG

Peak

Max Cap Check

Memory Check

DVD Check

Static IR Check

Power EM Check

Low Power Check

Voltage & Current w/f

Differential Voltage Check

Noise Coupling Check

Switch Id-sat Check

Switch Off State Check

CPM

GSR/Tcl Settings

Hot Spot Summary » DvD Check

DvD Check ?

- Dynamic Hot Spot Exploration
- Analysis Type : minWC
- Highlighting Top 3 DVD Hot spots in the design
- DVD Threshold used for PASS/FAIL : 100 mV

✖ Number of failing regions : 287/304 (94.4 %)

✖ Number of failing instances : 75489/75618 (99.8294 %)

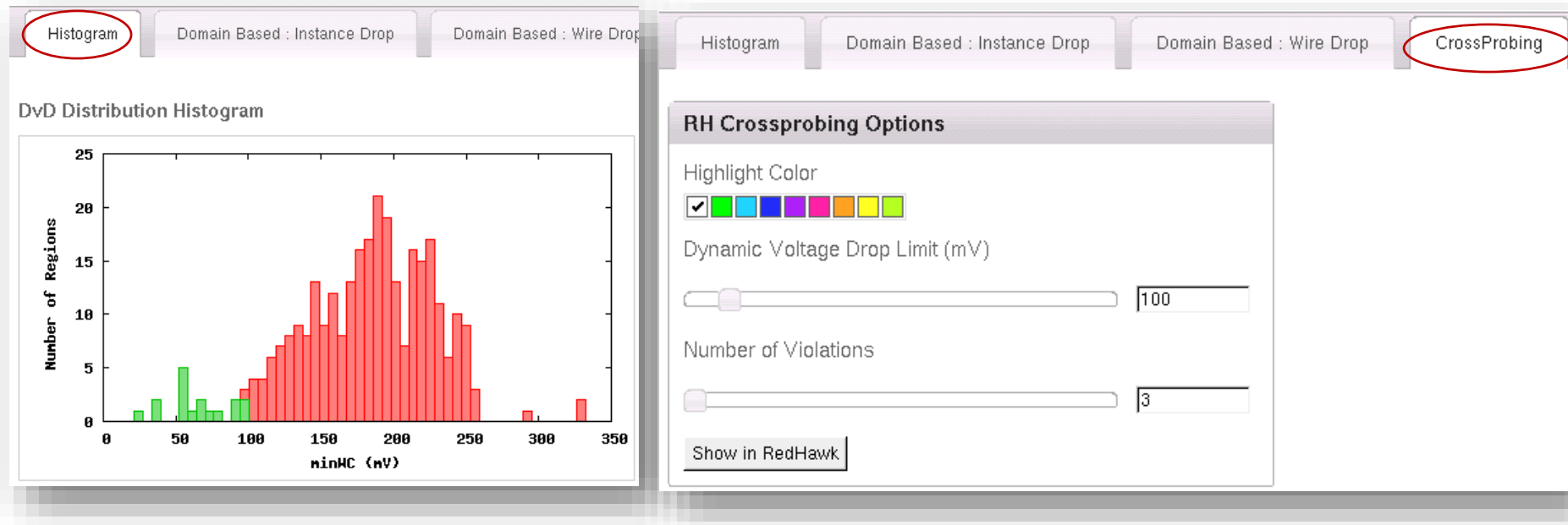
Top Violations

Region	Drop (mV)	Failed Instances	Status
🔍 Hot Spot : 1	335.01 (27.92 %)	1/1 (100.0 %)	✖
🔍 Hot Spot : 2	333.64 (27.80 %)	1/1 (100.0 %)	✖
🔍 Hot Spot : 3	298.09 (24.84 %)	1/1 (100.0 %)	✖



Show in RedHawk »

DvD Check : Summary



Histogram **Domain Based : Instance Drop** Domain Based : Wire Drop

Voltage Domain Based Hot spot Exploration

Domain	Failed Instances	Max Drop in mV(%)	Status
inst_129973/VDD_	37771/37771	237.22 (19.77 %)	✗
VDD	37718/37847	335.01 (27.92 %)	✗

Histogram Domain Based : Instance Drop **Domain Based : Wire Drop**

Voltage Domain Based Hot spot Exploration

Domain	Max Drop
VDD	178.337 mV
inst_129973/VDD_	125.095 mV

DVD Check

Checking Hot-spot #1

Top Violations

Region	Drop (mV)	Failed Instances	Status
Hot Spot : 1	335.01 (27.92 %)	1/1 (100.0 %)	✗
check-1	333.64 (27.80 %)	1/1 (100.0 %)	✗
Hot Spot : 3	298.09 (24.84 %)	1/1 (100.0 %)	✗

Hot Spot Summary » DvD Check » Hotspot : 1

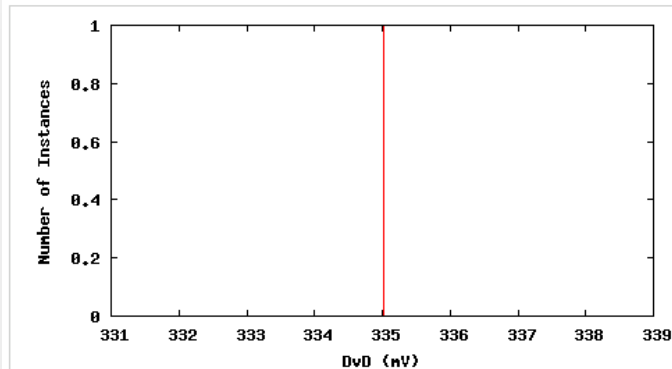
DvD Check

- Dynamic Hot Spot Exploration inside HOT SPOT : 1
- Hot spot region : 2304.64 2462.54 2592.46 2963.24
- Region id : 1281
- Instance Threshold used for PASS/FAIL : 100 mV

✗ Number of instances failing : 1/1 (100.0 %)
● Displaying Top 1 instances inside this HOT SPOT

Instance	DROP (mV)	Status
Hot Instance : 1	335.01 (27.92 %)	✗

Dynamic Drop Histogram inside Region : 1



Instance level histogram
within hot-spot 1

Detailed analysis inside the
region

HotSpot Map

DVD HOTSPOT MAP



Root Cause Identification

Design Weakness Analysis

Root Cause : Design Weakness

Root Cause : Data Integrity

Histogram

Root Cause : Design Weakness

Check	Status	Value	Ratio	Rank
POWER DISTRIBUTION QUALITY	✓	Total Power = 0.009012 W	1.10	108/662
PAD PLACEMENT QUALITY	✓	Current from nearest pad of Domain VSS : -14.35 mA Current from nearest pad of Domain VDD : 6.92 mA	NA	NA
CLOCK BUFFER CLUSTERING CHECK	NA	NA	NA	NA
PG RESISTANCE DISTRIBUTION	✓	Max Instance Resistance = 1.00 Ohm	1.00	160/662
PG RESISTANCE IMBALANCE	NA	NA	NA	NA
PEAK CURRENT DISTRIBUTION	✓	Total Peak Current = 46.38 mA	0.59	144/662
DECAP DISTRIBUTION	✗	Total Decap = 0.00 pF	0.00	35/662
PACKAGE DROP CONTRIBUTION	✓	Package Drop in Domain VSS : 10 mV (0 %) Package Drop in Domain VDD : 10 mV (0 %)	NA	NA
SIMULTANEOUS_SWITCHING_CHECK *	✓	Overall Switching Activity = 12288/80878 (15.19 %) Peak current/Average Current Ratio = 1181.5 mA/502.86 mA(2.34)	NA	NA

* These checks are not for this hot bucket but they belong to the entire design

There are no decaps in this region

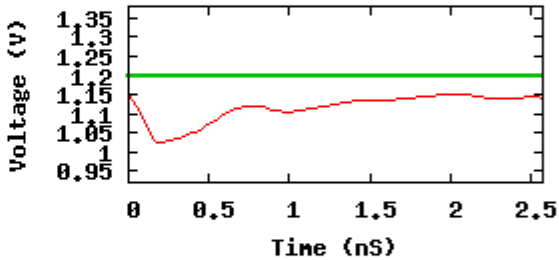
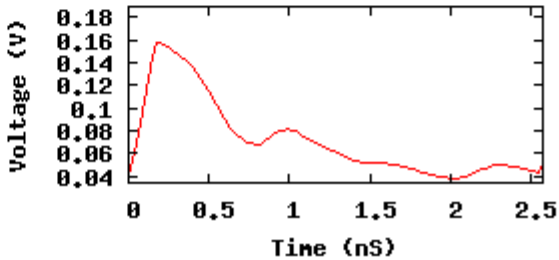
DVD Check

Instance Level Debug

Hot Spot Summary » DvD Check » Hotspot : 1 » Hot Instance : 1

DvD Check ?

- Dynamic Hot Spot Exploration for Hot Instance : 1
- Instance : inst_129422/inst_7778
- Instance Pin Threshold used for PASS/FAIL : 50.0 mV
- Pin Based Voltage Drop Details

Pin	Domain	Drop (mV)	Status	Pin DVD Waveform
VDD	VDD 	177.0 (14.75 %)	Fail	Plot for pin VDD 
VSS	VSS 	158.0 (13.16 %)	Fail	Plot for pin VSS 

Pin based voltage drop details
of the hot instance

DVD Check

Instance Level Debug

Properties	Data Integrity	Design Weakness
Instance Properties		
Cellname	cell_28	
Peak Current	VDD 46378.0 uA VSS 46378.0 uA	
Decap	0 pF	
Frequency	7.8125e+08 Hz	
Fanout	72	
Load	0.328 pF	
Switching Status	Switching	
Average Power	0.00901237 W	
Leakage Power	0 W	
Toggle Rate	1.00575	
Gridcheck Resistance	VDD Resistance : 9.68153 Ohm VSS Resistance : 9.73051 Ohm	

Properties	Data Integrity
Data Integrity	
Data	Status
Apl Current	✗
Apl Cap	✗
Apl Pwcap	NA
Lef	✓
Lib	✗
Sta	✓
Spef	✗
Gds2def	✗

Properties

Data Integrity

Design Weakness

Design Weakness

Check	Status	Value	Rank
Peak Current	✗	46378.0 uA	1/1
Resistance	✗	19.412 Ohm	1/1
Power	✗	0.00901237 W	1/1
Load	✗	0.328 pF	1/1

DVD Check

Instance Level Debug: SPT and Resistance Report

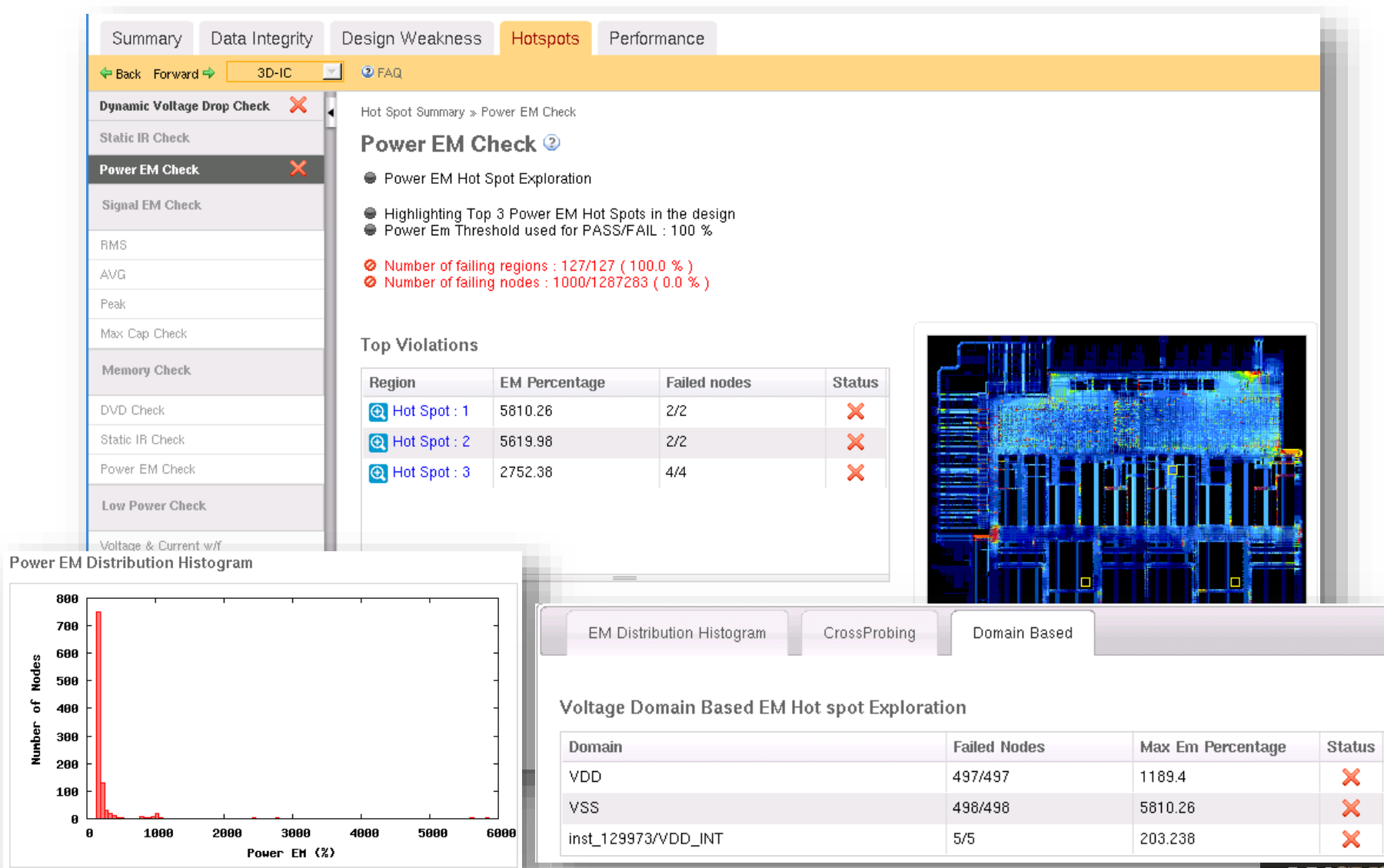


Equivalent Tcl Command:

perform min_res_path -o res_path.rpt

User can directly zoom into the bottleneck segments

Power EM Check : Summary



Power EM Check

Checking Hot-spot #1

Hot Spot Summary » Power EM Check » Hotspot : 1

Power EM Check ?

- Power EM Hot Spot Exploration inside HOT SPOT : 1
- Hot spot region : 3985.200000 1660.500000 4151.250000 1826.550000
- EM Threshold used for PASS/FAIL : 100 %

❌ Number of nodes failing : 2/2 (100.00 %)

- Displaying Top 1 nodes inside this HOT SPOT

Node	EM Percentage	Status
Hot Node : 1	5810.26	❌

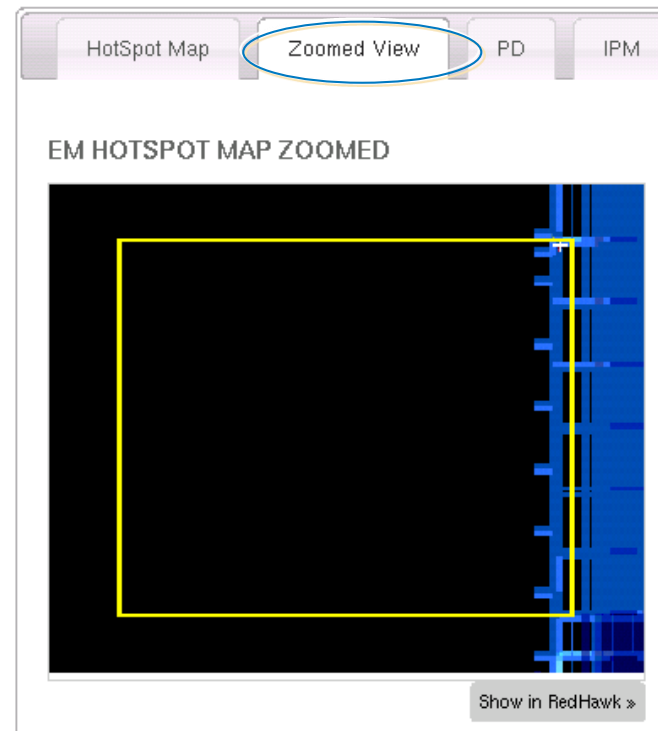
Hot Spot Summary » Power EM Check » Hotspot : 1 » Hot Node :

Power EM Check ?

- PowerEM Hot Spot Exploration for Hot node : 1
- Node : node1

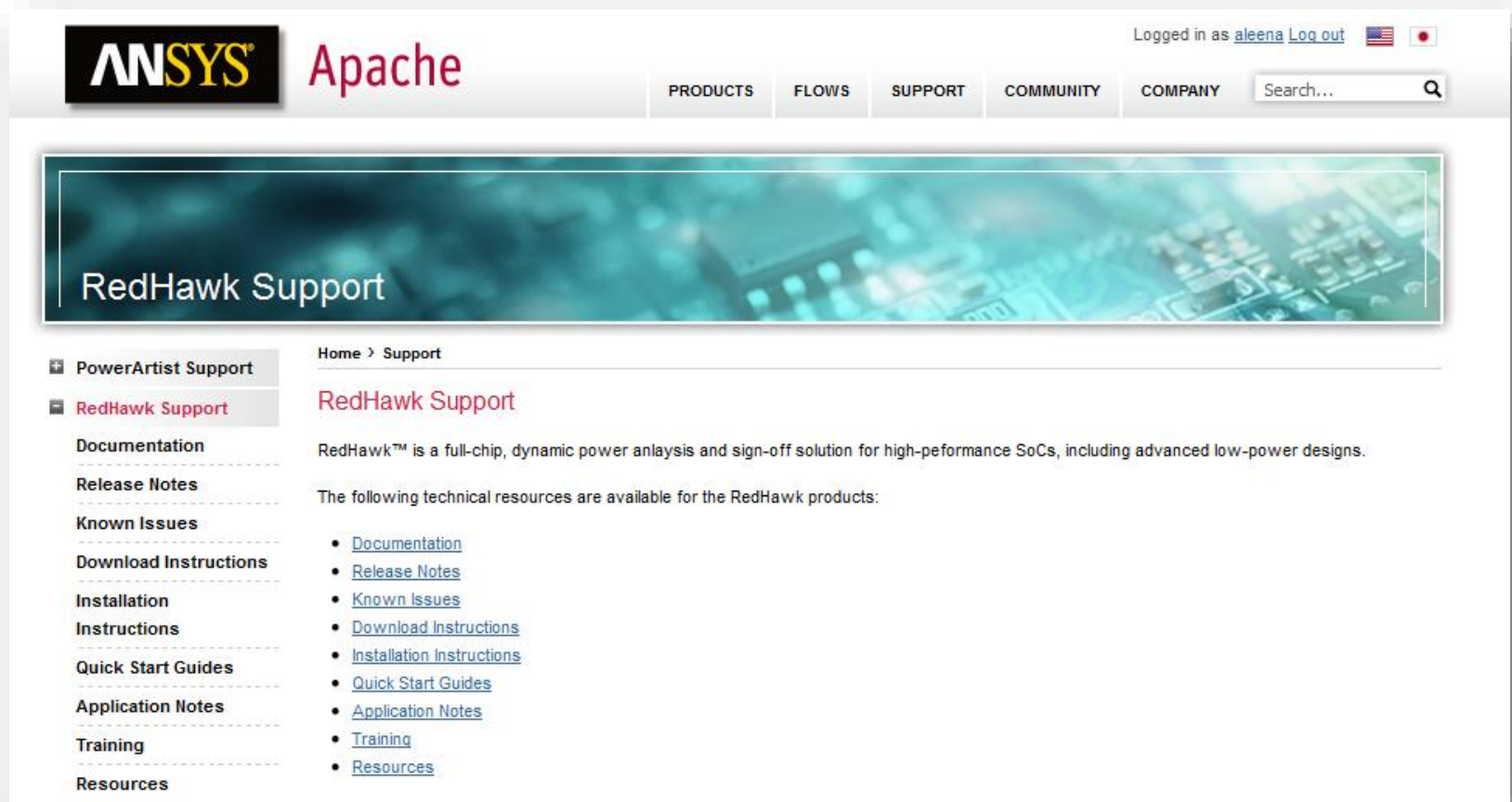
WIRE/VIA PROPERTIES

Location	4.148085e+03,1.824885e+0
Em percentage	5810.26 %
Layer	METAL4
Node Type	WIRE
Net	VSS
Width	2.000000e+00 um



How to get Help!!

- Apache Online Customer Support Center
 - <http://support.apache-da.com>
 - Email: support@apache-da.com



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Thank You!!!